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# AuthorNet: Leveraging attention-based early fusion of transformers for low-resource authorship attribution

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ABSTRACT

Authorship Attribution (AA) is crucial for identifying the author of a given text from a pool of suspects, especially with the widespread use of the internet and electronic devices. However, most AA research has primarily focused on high-resource languages like English, leaving low-resource languages such as Bengali relatively unexplored. Challenges faced in this domain include the absence of benchmark corpora, a lack of context-aware feature extractors, limited availability of tuned hyperparameters, and OOV issues. To address these challenges, this study introduces AuthorNet for authorship attribution using attention-based early fusion of transformer-based language models, i.e., concatenation of an embeddings output of two existing models that were fine-tuned. AuthorNet consists of three key modules: Feature extraction, Fine-tuning and selection of best-performing models, and Attention-based early fusion. To evaluate the performance of AuthorNet, a number of experiments using four benchmark corpora have been conducted. The results demonstrated exceptional accuracy: 98.86 ± 0.01%, 99.49 ± 0.01%, 97.91 ± 0.01%, and 99.87 ± 0.01% for four corpora. Notably, AuthorNet outperformed all foundation models, achieving accuracy improvements ranging from 0.24% to 2.92% across the four corpora.

## 1. Introduction

Authorship Attribution (AA) is an emerging research field of natural language processing and *forensic* author text analysis. Forensic text analysis is the process of examining the main attributes of a disputed text in order to conclude its original author. The key attributes are more particular clauses, specific tenses, distinguished sentence structure, or open and closed sentences with an appropriate grammatical constituent. These key attributes can be used to identify a disputed text's authorship. AA is a well-established research area concerning high-resource languages (HRLs) owning to availability of classification frameworks and linguistic corpora (Hossain & Hoque, 2023). However, AA is crucial for low-resource languages, e.g., the Bengali language, due to the shortage of context-aware text-to-feature extraction methods, the unavailability of well-tuned feature extraction and classification hyperparameters, and the lack of corpora, and the unavailability of domain-specific research guidelines. On the other hand, deep learning frameworks for high-resource languages with well-tuned hyperparameters could not achieve the best performance for low-resource languages like Bengali because of local dialects and morphological variations (Hossain et al., 2021b).

In recent years, AA has attracted much attention from Bengali research communities due to the widespread use of social media communication and the popularity of online learning systems. Social media textual information is spreading very quickly worldwide from individual and private or public group posts. Most of the text information's spread from the anonymous post without a fact check. This anonymity and misinformation increase the unrest, and security agencies cannot detect the suspects. AA will be an efficient solution to control the unrest situation and detect the actual suspect. There are many AA applications such as checking students' online submissions, analyzing fraudulent documents, detecting terrorist conspiracy texts, social media status updates to check the authenticity and identify fraud, plagiarism checks, controlling information crimes, and reducing textual forgeries on the internet. These applications are available for high-resource languages (e.g., English). Still, there are no usable Bengali language processing

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(BLP) tools or applications for Bengali authorship attribution which control textual forgeries on the internet. Therefore, these applications have motivated the development of the authorship attribution system, i.e., **the first motivation of this research**.

There are several traditional machine learning (ML) strategies for developing an AA model that deals with the extraction of different stylometric features from the text data and then feeding to the ML classifiers (Chakraborty, 2012; Das et al., 2015; Hossain et al., 2017; Luyckx et al., 2006). However, the stylometric features do not capture the semantic and syntactic attributes, and traditional ML classifiers (e.g., SVM & SGD) cannot manage the high dimensional texts (Hossain & Hoque, 2018). The character-embedding and n-gram features with traditional classifiers are the most commonly used methods for AA (Anisuzzaman & Salam, 2018; Islam et al., 2017; Khatun et al., 2019; Pal et al., 2017; Rakshit et al., 2015; Ruder et al., 2016; Tamboli & Prasad, 2013). Nevertheless, character level and n-gram feature extractors cannot represent the sentence and document level semantic and syntactic features. The word-level embedding (e.g., GloVe, FastText & Wor2Vec) and Deep CNN-based classifiers are the better-performing models for AA. Still, the word-level embedding only considers the local context and cannot overcome **OOV** issues (Borroto & Ricca, 2023). *Those are the **second motivation** for development of the Bengali authorship attribution system.*

Recently, transformers-based language models (e.g., M-BERT, ELECTRA & XLM-Roberta) and fine-tuning tasks, e.g., authorship attribution, achieved the State-Of-The-Art performance for high resource language (like English) (Suman et al., 2021). Whereas the Wu and Dredze (2020) proved that the multilingual transformer-based language models do not perform well for low-resource languages, and fine-tune models also failed to obtain better performance due to shortage of language model vocabularies, lower performance of Byte pair encoding (BPE) method, and lack of tuned hyperparameters of text-to-feature extraction and classification methods for Bengali. *These challenges of transformer-based models have also motivated the development of the authorship attribution system, i.e., **the third motivation**.*

The ***fourth motivation*** for incorporating both monolingual and multilingual feature fusion in Bengali authorship attribution is driven by the desire to leverage the unique advantages offered by each approach. Monolingual feature fusion allows us to tap into the intricate linguistic nuances and specific characteristics of the Bengali language, enabling us to capture fine-grained authorship patterns and linguistic features (Chen et al., 2023). On the other hand, multilingual feature fusion enables us to benefit from the broader knowledge and cross-lingual transferability inherent in multilingual transformer-based language models. By combining the insights and expertise acquired from both monolingual and multilingual models, we can achieve a more comprehensive and robust attribution system.

This research addresses several challenges in Bengali authorship attribution, including OOV issues, lack of well-tuned hyperparameters, and the shortage of domain-specific linguistic features in multilingual transformer-based language models. To overcome these challenges, the research introduces a system called AuthorNet based on an attention-based early fusion approach. AuthorNet combines the strengths of both multilingual and monolingual features by leveraging transformer-based language models, and integrating domain-specific knowledge through attention-based feature fusion. In summary, the study focuses on answering the following research questions **(RQs)** to provide insights into the proposed methodology and its effectiveness.

- **RQ1** How can transformer-based language models address out-of-vocabulary issues, and how does this solution impact authorship attribution tasks?
- **RQ2** How does an attention-based early fusion of transformer-based language models enhance low-resource authorship attribution performance?
- **RQ3** How do the model-level factors and specific components of AuthorNet, namely maximum sequence length, dropout value, number of attention layers, and attention layer embedding size, impact the performance of authorship attribution tasks?

The key contributions of this research and possible Answer to the Research Questions (**ARQs**) are summarized as follows:

- **ARQ1** Employed transformer-based language models for text-to-feature extraction, and the byte-pair encoding method is utilized to segment unknown words, enabling the construction of feature vectors for out-of-vocabulary (OOV) words (i.e., Section 6.1). The evaluation of the impact of OOV issues on authorship attribution tasks is presented in Section 8.2.6.
- **ARQ2** Identified the best-performed monolingual and multilingual transformer-based language models based on the performances of the fine-tuned models. These models are then utilized to construct an attention-based early fusion approach, as discussed in Sections 6.2 and 6.3.1. The addition of the attention layer and fusion operation effectively enhances the performance of authorship attribution, as outlined in Section [reference to the section discussing the performances of best-performed models] and Section 8.2.3.
- **ARQ3** We conduct ablation studies at both the model-level and AuthorNet component level to examine the impact of individual models, maximum sequence length, dropout value, number of attention layers, and attention layer embedding size on the performance of authorship attribution tasks, as discussed in Section 8.2.5.
- Empirically evaluated a total of 116 models that were generated by combining three non-contextual embeddings {GloVe, FastText, Word2Vec} with three deep classification models {CNN, LSTM, GRU}, eight transformer-based language models {Bangla-BERT, B-ELECTRA, M-BERT, XML-RoBERTa, DistilBERT, DeBERTa, Indic-BERT, MuRIL}, and four early fusion models {concatenation, averaging, concatenation+attention, averaging+attention}. These models were evaluated using four benchmark corpora named BACC-18, BADD16, BADD6, and LD. Further details can be found in Section 8.1, Table 4, Section 8.2.2, Table 5, and Section 8.2.3, Table 6.

The paper is structured as follows: Section 2 presents the related work, Section 3 describes the problem statement, and the details of the foundation models are provided in Section 4. The statistics of the corpora are illustrated in Section 5. The proposed AuthorNet is explained in Section 6. The experimental settings are described in Section 7, and the results and discussion are summarized in Section 8. Error analysis of the proposed and foundation models is presented in Section 9, while Section 8.3 discusses the significant findings of the proposed system. Finally, Section 10 presents the concluding remarks and future research directions.

## 2. Related work

Although authorship attribution is a well-known research problem for resource-available languages, it is a primitive stage for low-resource languages. This study addresses the problem of authorship attribution in a low-resource language, such as Bengali. Therefore, previous research on AA is broadly categorized into non-Bengali and Bengali languages.

### 2.1. AA on non-Bengali languages

There is an enormous amount of research has been conducted in non-Bengali languages (e.g., English, European, Arabic, Latin, and Urdu). The recent non-Bengali AA studies were grouped into two categories, e.g., Machine Learning (ML) based AA and Deep Learning

(DL) based AA. The notable and recent works from each category are described in the following.

**AA using ML-based Methods:** The ML-based non-Bengali AA methods have extracted the features using TF-IDF and statistical Stylometry. The statistical classifiers, e.g., Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), single layers Neural Networks (NN), and K-Nearest Neighbors (KNN), are used to produce the attribution models. For example, a Latin text language AA system has evolved in Kabala (2018) using Bray–Curtis distance and logistic regression for attribution and achieved maximum accuracy of 99.86%. The developed system does not consider the higher dimension data and does not consider the deep and transformers-based learning system. Anwar et al. (2018) introduced an unsupervised ML (e.g., LDA) based AA system for Urdu text language and achieved an $F_1$ score of 92% for 6,000 texts. Jambi et al. (2021) designed an Arabic short-microblog attribution system using SVM and KNN approaches. The classifier has attained maximum accuracy of 65.00% for 1,000 authors with 1.5 million Arabic tweets. The statistical classifiers (e.g., Bagging & AdaBoost) with ensemble method used for Arabic enquires attribution system (Al-Sarem et al., 2020). The developed method used the stylometric features and evaluated the self-build dataset. Taha et al. (2021) explored a two-level AA learning system for Arabic author texts. The developed system initially extracts the vocabulary level feature and fed to the second learning. The vocabulary level statistical feature does not capture the word semantics. Anwar et al. (2018) evolved a word n-grams of LDA method for Urdu AA that obtained 92.00% $F_1$-score on 6,000 Urdu Newspaper articles. The stylometry feature-based AA has developed in Tweedie et al. (1996) for the English language using neural networks (NN) and clustering frameworks. The stylometry features cannot capture the word-level semantic and syntactic attributes, and NN and clustering classifiers do not represent the sentence-level global attribution features.

**AA using DL-based Methods:** Suman et al. (2021) developed a AA system using the capsule-based CNN method for English text. The developed method used the GloVe model for feature extraction and capsule-based CNN for attribution. The GloVe model only considers the word level non-context feature. A character-based embedding for features extraction and multi-channel CNN for attribution is used in Ruder et al. (2016) with the English author texts. The character embedding takes huge amount of memory for embedding model training and does not consider the syntactic features. Rocha et al. (2017) explored an English AA system for forensic text analysis that studied different deep learning techniques. The developed system is considered for short texts but does not experiment with long text documents. The backpropagation-based particle swarm operation is applied to identify the Java source code attribution system and obtained 91.06% accuracy for 2,022 Java source code (Yang et al., 2017). Bander et al. (2017) also developed a system for source code attributions system with 200 source code from 10 programmers and gained 85.00% accuracy using an LSTM-based classifier. Fabien et al. (2020) investigated the transformer-based AA for English IMDb, Enron, and Blog datasets. However, most of the previous research has been conducted for non-Bengali AA, and none of this research considers the hyperparameters optimization techniques. Thus, we adapted hyperparameters of foundation models for Bengali authorship attribution systems.

### 2.2. AA on the Bengali language

The high-resource language (e.g., English) AA and forensic author text analysis have been feasible because of the availability of linguistic resources and well-tuned feature and classification methods. However, Bengali authors' attribution, identification, profiling, and forensic analysis are relatively primitive research issues for the low-resource domain. Recently, some research has been conducted on Bengali AA based on machine learning (ML) and Deep learning (DL). The remarkable and recent studies from each category are described in the following.

**AA using ML-based Methods:** There is a much number of Bengali AA studies that have been conducted using ML-based methods. For example, Lahiri et al. (2017) explored a statistical machine learning-based (e.g., SVM & Random Forest) technique for three Bengali authors of 3,000 texts. They used the n-grams text feature and Random Forest (RF) classification method, which obtained 98.00% test accuracy but experimented with only three authors self-build the dataset. Lahiri et al. (2016) used supervised learning, e.g., multi-layer perception (MLP), for authorship attribution that obtained maximum accuracy of 99.00%. However, this study experimented on three Bengali author datasets. A stylometric feature and SVM-based classifiers were used in Das et al. (2015) for Bengali AA but have not considered the word and sentence level features. Chakraborty (2012) introduced the statistical classifiers for Bengali authorship identification and attained maximum accuracy of 83.3% using the SVM classifier. Tamboli and Prasad (2013) applied different statistical methods in Bengali AA. They claimed that n-gram-based features gained 90.00% accuracy using a self-built dataset. A stylometry feature extraction and ML-based ensemble achieved an accuracy of 90.67% for 700 blog articles (Hossain et al., 2017). Anisuzzaman and Salam (2018) developed a three Bengali author dataset containing 107,380 words, and Naive Bayes (NB) method obtained 95.00% accuracy. Pal et al. (2017) explored SVM and NB methods for Bengali AA. The SVM classifier has the best accuracy of 90.74% for 20 Bengali bloggers datasets. A Bengali poetry identification method is developed in Rakshit et al. (2015) using multi-class SVM classifier and achieved 92.00% identification accuracy for 4 authors and 1058 poems. The poetry identification method experimented on a limited number of poems (e.g., 1058). Islam et al. (2017) introduced a Bengali AA method using NB and RF, which achieved a maximum accuracy of 96.00% for the RF classifier. Most previous works have been revealed based on stylometric features and statistical classification techniques, e.g., SVM, RF, NB, and MLP. However, these cannot handle large volumes of training data and cannot extract the word and sentence level semantic and syntactic features.

**AA using DL-based Methods:** In recent times, DL-based studies have been conducted on the Bengali AA to solve the ML-based limitations. Khatun et al. (2019) explored the AA method using character-level embedding with CNN. Their developed method obtained maximum accuracies of 98.00% and 81.2%, respectively, for BAAD6 and BAAD14 datasets. The character-level embedding method failed to carry out the word-level semantic and syntactic information, whereas their CNN hyperparameters are not tuned. The fusion-based, e.g., CNN and artificial neural networks (ANN) Bengali AA system has been introduced in Dipongkor et al. (2021). They claimed the maximum accuracy of 80.74% for Bengali-Quora corpus (BQC) using the ANN method, 96.71% for BAAD6 using CNN+ANN and 99.78% for BAAD16 using the ANN method. Their classification frameworks (e.g., ANN & CNN)hyperparameters are not tuned for the Bengali corpus. A frequency-based GloVe with tuned CNN (e.g., GloVe+CNN) method is used in Hossain et al. (2021a). The GloVe+CNN method achieved maximum accuracies of 93.45%, 95.02%, and 98.67% for the datasets BACC-18, BAAD16, and LD, respectively. However, the CNN model manual hyperparameters tuning has taken enormous time and memory space. Chowdhury et al. (2018) represented a comparative analysis of several word embedding (e.g., GloVe, FastText & Word2Vec) and classifiers (e.g., ANN, RNN & CNN). They achieved a maximum accuracy of 86.95% for six authors using the GloVe+CNN model. The GloVe+CNN model did not consider the OOV issues and used the hyperparameters without tuning. Alam et al. (2020) introduced the Bengali AA using transformer-based frameworks and obtained a maximum accuracy of 93.41% for BAAD16 using the XLM-Roberta-large fine-tuned model.

*Differences from existing AA researches.* Previous Bengali authorship attribution studies primarily relied on non-contextual feature extraction methods such as TF-IDF, N-Gram, GloVe, FastText, Word2Vec, and character embedding. These approaches overlooked the issues of OOV

words and lacked context awareness, limiting their ability to capture semantic and syntactic meaning (Mutlu & Sezer, 2023). Additionally, the default hyperparameters used in attribution models failed to achieve optimal performance for the low-resource languages. To address these limitations, the proposed approach, called AuthorNet, introduces several key advancements. ***Firstly***, it leverages transformer-based language models for text-to-feature extraction, allowing for the incorporation of context-aware features and effectively addressing OOV issues. ***Secondly***, it introduces an hyperparameters tuning method specifically designed for low-resource authorship corpora. ***Thirdly***, it utilizes an attention-based early fusion of transformers, combining the strengths of both monolingual and multilingual models (Gheewala et al., 2024). ***Lastly***, it evaluates the attribution performance across four corpora, investigating the impact of k-fold cross-validation and comparing the performance of monolingual and multilingual models in terms of vocabulary size. These innovative contributions distinguish our research from existing forms. In contrast, prior approaches heavily relied on non-contextual feature extraction, grappled with challenges related to out-of-vocabulary (OOV) words, needed more context awareness, and struggled with default hyperparameters. To the best of our knowledge, no previous research has addressed these issues comprehensively, making AuthorNet a novel and promising approach in the field of low-resource authorship attribution.

## 3. Problem statement

The main problem of this research is developing an authorship attribution system using a predefined corpora $C_i$, where $C_i =< \mathcal{T}_i, \mathcal{A}_i >$. The $\mathcal{T}_i$ denotes the text space, $\mathcal{A}_i$ denotes the corresponding author space where $i = [1, 2, \dots, N']$, and $N'$ represents the total number of corpora (Rocha et al., 2017; Theophilo & Rocha, 2022). To articulate this problem, we introduce AuthorNet, a system for authorship attribution. AuthorNet encompasses three key phases: (i) Feature extraction, (ii) Fine-tuning and selection of best-performing models, and (iii) Attention-based early fusion. The subsequent sections provide a brief overview of each of these phases:

*Feature extraction.* In this stage, the inputs consist of the $j$th sample from the $i$th corpus $C_i^j$ and the transformer-based language model $TL_l$. The output is the embedding feature $\mathcal{F}il^j$ derived from $Ci$ using Eq. (1).

$$\mathcal{F}_{il}^j = \Phi_l^{LM}(C_i^j, TL_l), j = 1, \dots, L; i = [1, 2, \dots, N']; l = [1, 2, .., M'] \quad (1)$$

Where $\Phi_l^{LM}(.,)$ denotes the feature extraction function. The term $M'$ denotes the total number of transformer-based language models, and the specifics of feature extraction are elaborated in Section 6.1.

*Fine-tuning and selection of best-performing models.* In the second phase, inputs are the transformer-based language model $TL_{il}$ with tuned hyperparameters of foundation models $\mathcal{H}_l^t$ and feature matrix $\mathcal{F}_{il}^j$. The output is the fine-tuned attribution model $TL_{il}^{FT}$. This fine-tuned language model is trained using Eq. (2).

$$TL_{il}^{FT} = \Phi_l^{FT}(TL_{il}(\mathcal{H}_l^t), \mathcal{F}_{il}^j) \quad (2)$$

The parameters $sl$ and $D_f$ represent the maximum sequence length and the corresponding dimension for features. Among these models, the two best-performing models $TL_z^{FT}, z = TL^{best1}, TL^{best2}$ are selected based on statistical significance tests for attention-based early fusion training and inferencing. Fine-tuning and models selection are elaborated in Section 6.2.

*Attention-based early fusion.* In the third phase, an attention-based early fusion of two best-performing transformer-based language models $TL^{best1}$ and $TL^{best2}$ are fused to generate the final authorship attribution model $\Theta_i$ using Eq. (3).

$$\Theta_i = \Gamma(TL_z^{FT}, \Phi_z^{LM}(C_i^j)), z = \{TL^{best1}, TL^{best2}\} \quad (3)$$

**Table 1**
Embedding frameworks and hyperparameters summary.

| Frameworks | Optimized hyperparameters |
|---|---|
| GloVe | CW: 12, ED: 300, MW: 2, MI: 30 |
| FastText | CW: 13, ED: 250:, MW: 2, MI: 30, model: sg |
| Word2Vec | CW: 15, ED: 300:, MW: 2, MI: 30, model: sg |

Here $\Gamma(.,)$ is the concatenation with an attention-based early fusion function. This model $\Theta_i$ is used for unknown author attribute (i.e., author). Finally, the attribution model $\Theta_i$ predict the $k$th unknown sample (test-sample) for corpus $i$ using Eq. (4).

$$\mathcal{A}_i^k = \Psi(\Theta_i, \Phi_z^{LM}(C_i^k)), k = 1, \dots, M \quad (4)$$

Here, $\Psi(.,)$ is the prediction function, and in this experiment, a softmax function is used as the prediction function. $\mathcal{A}_i^k$ is the predicted attribute vector for $i$th corpus. The $k$th unknown sample text is defined as $C_i^k \subset C_i$ and $M$ indicates the total number of test samples in $C_i$. The details are described in Section 6.3.

## 4. Foundation models for the authorship attribution

In this section, we have discussed the feature extraction and authorship attribution foundation models that serve as a comparison for evaluating the performance of AuthorNet. Table 1 presents an overview of the embedding frameworks and their respective hyperparameters utilized for feature extraction in authorship attribution tasks. Table 1 hyperparameters CW, ED, MW, MI, and model indicate the contextual windows (CW), embedding dimension (ED), minimum frequency word count (MW), the maximum number of epochs (MI), and model variant (model), respectively. These parameters are used at the embedding frameworks training phase and optimized using embedding parameters identification (EPI) Algorithm (Hossain et al., 2021b). The GloVe is *frequency-based* non-contextual feature extractor framework, whereas Word2Vec and FastText are the *prediction-based* frameworks. The Skip-gram variation of FastText and Word2Vec is used in this research.

Table 2 shows the authorship attribution frameworks and hyperparameters. The LSTM, GRU and CNN frameworks hyperparameters are taken from Theophilo et al. (2023). The sequence length indicates the maximum number of words used in the attribution model during the training phase. The single-layer multi-kernel CNN takes 200 epochs to converge the model during training. The transformer-based fine-tuning models start with these initial hyperparameters and optimize using the proposed hyperparameters tuning (HT) Algorithm. The three initial hyperparameters, e.g., batch size are {6, 8, 12} where initial epochs {10, 15, 20} and maximum sequence lengths are {100 250, 400} are used in the fine-tuning phase.

The Bangla-BERT (Bhattacharjee et al., 2021), and Bengali-ELECTRA (B-ELECTRA)[1] language models are trained only for the Bengali language, whereas Indic-BERT (Kakwani et al., 2020), and MuRIL (Khanuja et al., 2021) are trained in 13 and 17 Indian languages, including Bengali. The M-BERT (Devlin et al., 2019) and DistilBERT (Sanh et al., 2019) are trained for 104 multiple languages, including Bengali. The XLM-Roberta (Conneau et al., 2020) and DeBERTa (He et al., 2021) language models trained for 100 languages, including Bengali.

## 5. Corpora

To evaluate the AuthorNet performance, this study used four Bengali authorship attribution corpora $C_i, i \in \{LD, BACC-18, BAAD16, BAAD6\}$.

[1] https://huggingface.co/monsoon-nlp/bangla-electra

**Table 2**
Foundation models optimized hyperparameters summary.

| Frameworks | Optimized hyperparameters |
|---|---|
| LSTM | layer: 2, sequence length: 400, hidden dimension: 256, 128, batch size: 32, dropout: 0.45, 0.48, loss: categorical_crossentropy, optimizer: adam, activation: tanh & softmax, epoch: 40 |
| GRU | layer: 2, sequence length: 400, hidden dimension: 256, 128, batch size: 64, dropout: 0.53, 0.57, loss: categorical_crossentropy, optimizer: adam, activation: LeakyReLU & softmax, epoch: 40 |
| CNN | layer: 1, kernel: 3, kernel size: [2,3,4], #kernel: 128, epoch: 200, dropout:0.54, sequence length: 400, activation: LeakyReLU & softmax, pooling: max |
| Bangla-BERT | layer: 12, attention heads: 12, vocab size: 102025, #language: 1, output dimension: 768 |
| B-ELECTRA | layer: 12, attention heads: 4, vocab size: 29898, #language: 1, output dimension: 256 |
| M-BERT | layer: 12, attention heads: 12, vocab size: 105879, #language: 104, output dimension: 768 |
| DistilBERT | layer: 6, attention heads: 12, vocab size: 119547, #language: 104, output dimension: 768 |
| Indic-BERT | layer: 12, attention heads: 12, vocab size: 200000, #language: 12, output dimension: 768 |
| MuRIL | layer: 12, attention heads: 12, vocab size: 197285, #language: 17, output dimension: 768 |
| XLM-Roberta | layer: 12, attention heads: 12, vocab size: 250002, #language: 100, output dimension: 768 |
| mDeBERTa-V3 | vocab_size: 251000, #language: 100, attention heads: 12, num_hidden_layers: 12, output dimension: 768 |

**Table 3**
Statistics of authorship classification Corpora.

| Attributes | BAAD16 | LD | BACC-18 | BAAD6 |
|---|---|---|---|---|
| #Author texts | 17,966 | 3,000 | 25,749 | 2,100 |
| #Author | 16 | 3 | 18 | 6 |
| Corpora size (MB) | 210.8 | 21.6 | 160.1 | 37.1 |
| #Words (M) | 13.47 | 1.25 | 9.86 | 2.30 |
| #Unique words (M) | 0.59 | 0.08 | 0.38 | 0.17 |
| #sentences (M) | 0.71 | 0.08 | 0.75 | – |
| #Tran author text | 14,374 | 2,250 | 22,894 | 1,800 |
| #Test author text | 3,592 | 750 | 28,88 | 300 |
| Max./Min. train author text size in KB | 16.3/9.6 | 31.4/1.4 | 30.9/0.7 | 222.75/0.05 |
| Max./Min. test author text size in KB | 14.6/10.0 | 37.5/1.2 | 118.5/0.8 | 95.56/0.05 |

**LD:** The LD is the Bengali literature corpus and contains three prominent Bengali authors Shanta et al. (2015). A total of 3000 literature texts split into 2,250 training texts and 750 texting texts, e.g., $C_1 =< \mathcal{T}_1, \mathcal{A}_1 >$ and $\mathcal{T}_1 = [1, \dots, 3000]$, $\mathcal{A}_1 = [1, \dots, 3]$). The corpus summary is shown in Table 3. Author (1) follows the Bengali Sadhu-bhasha linguistic variation and a few words taken from Sanskrit-bhasha. At the same time, authors (2) and (3) follow the Sadhu-bhasha and Cholito-bhasha linguistic variations.

**BACC-18:** BACC-18 is the largest corpus in terms of the number of authors. The corpus contains a total of 25,749 with 18 famous Bengali authors, e.g., $C_2 =< \mathcal{T}_2, \mathcal{A}_2 >$ and $\mathcal{T}_2 = [1, \dots, 25749]$, $\mathcal{A}_2 = [1, \dots, 18]$. The author's texts are collected from two Bengali domains, e.g., online literature and Blogs. Hossain et al. (2021a). The BACC-18 is a morphologically rich corpus because three Bengali linguistics variations (Sanskrit-bhasha, Cholito-bhasha & Sadhu-bhasha) are available in this corpus. The corpus contains author texts from $the18th$ to $20th$ decades. The overall corpus summary is shown in Table 3.

**BAAD16:** The BAAD16 is a Bengali multi-domain corpus (e.g., Novels, Stories, Series, and Blogs) and contains 16 different authors. The authors writing style follows the Bengali Cholito-bhasha and Sadhu-bhasha linguistic variations. Each of the texts contains 750 words and 14,374 training texts, and 3,592 inferencing texts, e.g., $C_3 =< \mathcal{T}_3, \mathcal{A}_3 >$ and $\mathcal{T}_3 = [1, \dots, 17966]$, $\mathcal{A}_3 = [1, \dots, 16]$. Table 3 shows the BAAD16 corpus summary.

**BAAD6:** The BAAD6 is the Bengali Blog-based corpus and contains six bloggers texts (Chowdhury et al., 2018). Most of the context of texts are religion, politics, sports, education, and lifestyle. There are 1800 training and 300 inferencing texts, e.g., $C_4 =< \mathcal{T}_4, \mathcal{A}_4 >$ and $\mathcal{T}_4 = [1, \dots, 2100]$, $\mathcal{A}_4 = [1, \dots, 6]$. Due to copyright and privacy issues, the corpus authors' actual names are not exposed.

## 6. Methodology of AuthorNet

The main goal of this research is to develop a authorship attribution system (i.e., AuthorNet) that can classify a piece of Bengali author text based on a predefined author name. The AuthorNet comprises three main modules, i.e., (i) Feature extraction (ii) Fine-tuning and selection of best-performing models, and (iii) Attention-based early fusion. Fig. 1 shows the abstract view of the AuthorNet system. Each of the modules is described in the following subsections.

### 6.1. Feature extraction

To address **RQ1**, this study leverages transformer-based language models to extract text features and effectively overcome the challenges associated with OOV words. During the text-to-feature extraction phase, the input comprises a collection of author texts, while the output consists of sets of 2D feature matrices. The $j$th training sample of $i$th corpus $C_i^j$ generates a feature matrix $F_{il}^j$, $F_{il}^j \in \mathbb{R}^{(sl \times D_f)}$ using

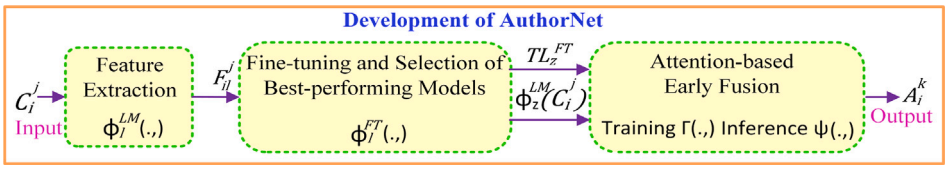



**Fig. 1.** Abstract view of the development of AuthorNet.

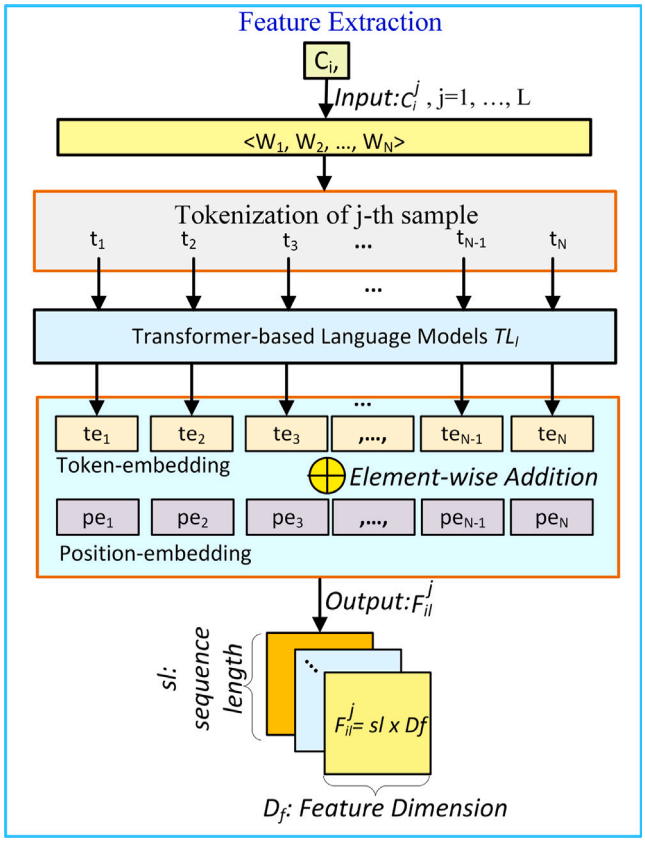



**Fig. 2.** Overview of text-to-feature extraction.

$l$th transformer-based language model. Where $sl$ represents the maximum sequence length and $D_f$ represents the feature dimension of $j$th test sample. Fig. 2 shows the details of the author's text to feature extraction. Each of the text is initially converted to a set of tokens, i.e., $\{t_1, t_2, \ldots, t_N\}$, where $t_N$ denotes the $N$th token of the given input text $C_i^j$. This set of tokens is passed through the transformer-based language model, and each token is converted to a numerical feature vector, i.e., token embedding using Eq. (1). The $\{te_1, te_2, \ldots, te_N\}$ indicates the token embedding set and $te_N \in \mathbb{R}^{D_f}$ represent the $N$th token embedding of $D_f$ feature dimension. To improve the word–word correlation, the position embedding set $\{pe_1, pe_2, \ldots, pe_N\}$ is element-wise added with the token embedding set, i.e., $\{te_1, te_2, \ldots, te_N\} \oplus \{pe_1, pe_2, \ldots, pe_N\}$ where $\oplus$ indicates the element-wise addition. Finally, the input text $C_i^j$ is converted to a 2D feature matrix, i.e., $F_{il}^j$, which is used for the fine-tuning phase.

### 6.2. Fine-tuning and selection of best-performing models

The fine-tuning module takes a set of feature matrices $[F_{il}^j]_{j=1}^{j=L}$ as input and generates two best-performing attribution models namely $TL_z^{FT}, z = \{TL^{best1}, TL^{best2}\}$. Here, $i$ represents the corpus $i$, $z$ indicates the best-performing language model $z$ and $L$ denotes the total number of training samples in the corpus.

The input feature matrix is passed through multiple attention heads. Each head calculates three attention vectors: query ($Q_{il} \in \mathbb{R}^{D_f}$), key ($K_{il} \in \mathbb{R}^{D_f}$), and value ($V_{il} \in \mathbb{R}^{D_f}$) using trainable attention matrices: query weight matrix $WQ_{il} \in \mathbb{R}^{(D_f \times D_f)}$, key weight matrix $WK_{il} \in \mathbb{R}^{(D_f \times D_f)}$, and value weight matrix $WV_{il} \in \mathbb{R}^{(D_f \times D_f)}$, respectively. These attention vectors, along with the feature matrix $F_{il}^j$, are passed through layer normalization and feed-forward network blocks, ultimately producing the classifiable feature vector $CLS \in \mathbb{R}^{D_f}$.

The classifiable feature vector is then fed into a Multilayer Perceptron ($MLP$) block, which generates the output corresponding to the expected author name from a predefined set of authors. The error value is calculated based on the output and used to update the trainable weight matrices through backpropagation. Consequently, the corpus $i$ produces the authorship attribution model $TL_{il}^{FT}$ using the transformer-based language model $l$. Among the eight fine-tuned attribution models, the attention-based early fusion training and inferencing module selects the best-performing model $TL_z^{FT}, z = \{TL^{best1}, TL^{best2}\}$. This study, based on the accuracy and statistical significance tests (i.e., Details in Section 8.2.2), the first best-performing model is MuRIL (i.e., $TL^{best1}$=MuRIL) and second best-performing model is Bangla-BERT (i.e., $TL^{best2}$=Bangla-BERT). This selection process is depicted in Fig. 1, and the results are summarized in Table 5. These two models, $TL^{best1}$ = (MuRIL) and $TL^{best2}$ = (Bangla-BERT), are subsequently utilized in the attention-based early fusion training phase.

### 6.3. Attention-based early fusion

This section initially trains an attention-based early fusion model and then infers the model. The following subsection describes each of them.

#### 6.3.1. Training attention-based early fusion

To address **RQ2**, this module introduces an attention-based early fusion approach that combines the two best-performing models $TL_z^{FT}, z = \{TL^{best1}, TL^{best2}\}$. The feature matrices of these models are represented as $F_{iz}^j$, and $j$ represents the $j$th test sample from the corpus $i$. The output of this module is an attention-based early fusion authorship attribution model, referred to as $\Theta_i$. The attention-based early fusion training and inferencing module for authorship attribution is visually depicted in Fig. 3.

In Fig. 3, the $j$th training sample of the corpus $i$ is represented by the MuRIL-based feature matrix denoted as $F_{iz(1)|z(1)=TL^{best1}}^j$, and the Bangla-BERT based feature matrix, denoted as $F_{iz(2)|z(2)=TL^{best2}}^j$. The $TL^{best1}$ feature encoding is represented by $CLS_1$, $P_{11}$, $\ldots$, $P_{1N}$, and $ESP_1$, while the $TL^{best2}$ based feature encoding is represented by $CLS_2$, $P_{21}$, $\ldots$, $P_{2N}$, and $ESP_2$. Here, $CLS_1$ and $CLS_2$ refer to single-dimensional vectors. Similarly, $P_{1N}$ and $P_{2N}$ represent the $N$th token feature vectors of $TL^{best1}$ and $TL^{best2}$ models, respectively, where $(P_{1N}, P_{2N})$. The $CLS_1$ and $CLS_2$ serve as document-level stylometric feature representations for the $j$th sample and are forwarded to the attention-based early fusion module. These two document-level feature vectors are fused using the concatenation operation, denoted as $CLS_1 \oplus CLS_2$. The concatenated feature vector is then passed through the multi-head attention block. The attention block calculates the attention value using Eq. (5).

$$A_{iz}^V = \left(\frac{CLS_{iz}^Q \times CLS_{iz}^K}{\sqrt{dim(CLS_z)}}\right) \times CLS_z \tag{5}$$

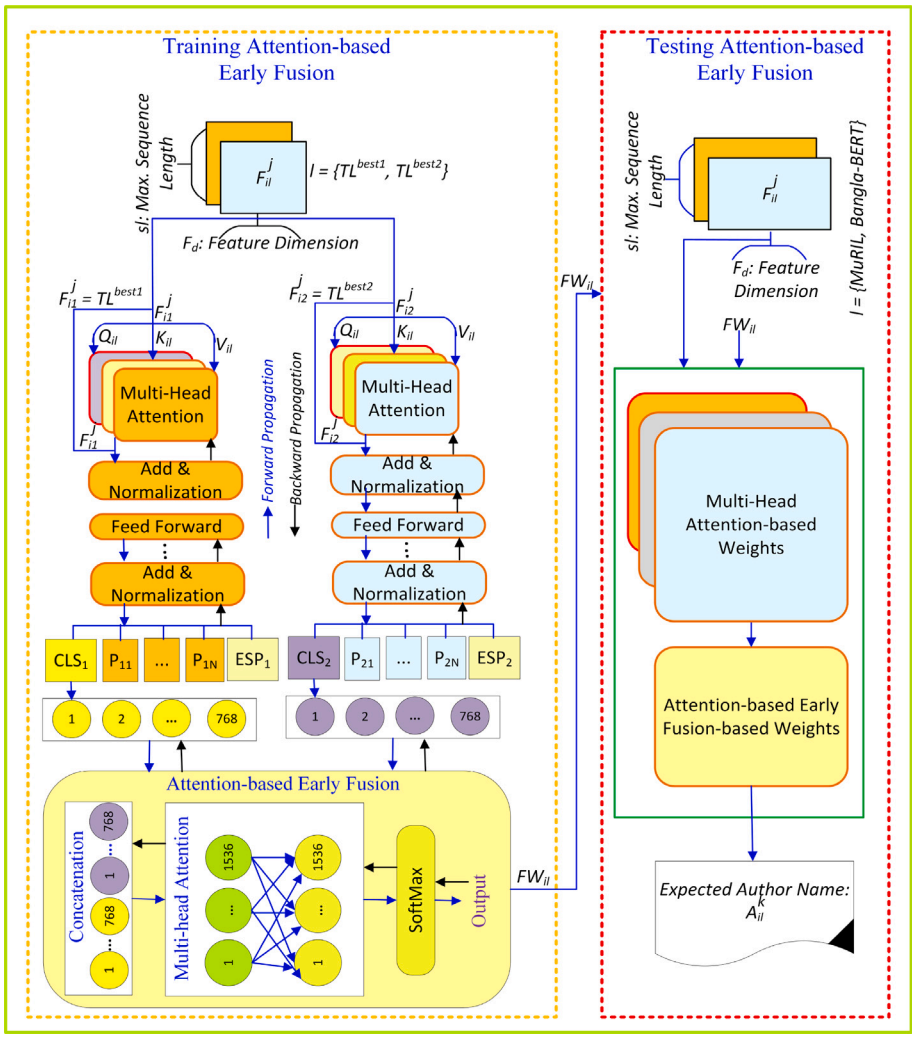



**Fig. 3.** AuthorNet: Overview of attention-based early fusion of authorship attribution training and inferencing.

In this context, $A_{iz}^{V}$ represents the attention-based vector, where the length of $A_{iz}^{V}$ is *1536*. $CLS_{iz}^{Q}$ and $CLS_{iz}^{K}$ refer to the model $z$'s trainable query and key vectors, respectively. $CLS_z$ denotes the encoded vector of the model $z$, and the dimension of $CLS_z$ is denoted as $dim(CLS_z)$. The $A_{iz}^{V}$ vector is normalized using the ReLU function and passed to the subsequent fully-connected layer. The attention-based early fused vector is then propagated through the $SoftMax$ layer to generate author-wise expected values. Deviations or errors from these values are calculated, and the model weights are adjusted through the backward pass operation. Finally, the fusion-based authorship attribution model $\Theta_i$ is forwarded to the inference of the Attention-based early fusion module.

### 6.4. Inferencing by attention-based early fusion

This module takes two inputs, i.e., a set of test samples $[C_i^k]_{k=1}^{k=M}$ and an attention-based early fusion model, $\Theta_i$ and produces one output, i.e., expected author name $A_i^k$. Here $M$ denotes the total number of test samples in the corpus $i$, and $z$ indicates the transformer-based model $z$. The $k$th test feature matrix $F_{iz}^k$ is produced by the function $\Phi_z^{LM}(.,)$ for the test sample $C_i^k$ and fed to the attribution model $\Theta_i$ that generates an expected author list $A_i^k$. This author list is forwarded to the $softMax$ block, which generates the expected author name $A_i^k$ from the predefined author list based on the maximum expected value. In this way, the corpus $i$ measures the performances of the attention-based early fusion fine-tuned transformer-based attribution model.

## 7. Experiments

In this study, we aimed to develop a model (**AuthorNet**) for Bengali authorship attribution tasks using various language models implemented by the Python-based packages from the PyTorch framework. We compared the proposed model with several foundation methods, including Convolutional (CNN), Sequential (LSTM and GRU), and Transformer-based language models (Bangla-BERT, B-ELECTRA, M-BERT, XLM-RoBerTa, DistilBERT, DeBERTa, Indic-BERT, and MuRIL). We implemented the proposed model and the foundation methods using PyTorch 1.9.0, Pandas, Sklearn libraries, Python3 (version: 3.6), Numpy, and Transformer (version: 4.9.0). The hardware configurations used in this study included a multi-core processor (core-i7) with NVIDIA RTX 3050 GPU (Internal GPU memory 8 GB) and 32 GB physical memory.

To evaluate the performance of the proposed and foundation models, we used several benchmark evaluation measures, including accuracy (A), precision (P), recall (R), a weighted average of precision ($mA_p$), a weighted average of recall ($mA_r$), macro-average precision ($mA_p$), macro-average recall ($mA_r$), F1-score (F1), Welch's t-test, and ablation analysis on the four corpora test sets (Prabhakar & Lee, 2023).

## 8. Results and discussion

In this section, we present the performance evaluation of the foundation method and AuthorNet using four different Bengali corpora,

**Table 4**
Best-performed foundation methods Accuracy (A) and F1-score (F1) of four corpora, Bold values indicate the best-performed models.

| Model | BACC-18(%) | | BAAD16(%) | | LD(%) | | BAAD6(%) | |
|---|---|---|---|---|---|---|---|---|
| | A | F1 | A | F1 | A | F1 | A | F1 |
| **CNN+GloVe** | **93.42** | **93.00** | **95.03** | **95.00** | **98.63** | **99.00** | **94.30** | **93.75** |
| LSTM+GloVe | 91.03 | 90.78 | 92.15 | 91.00 | 97.05 | 97.00 | 92.90 | 93.00 |
| GRU+GloVe | 90.87 | 91.00 | 91.59 | 92.00 | 97.12 | 97.00 | 91.21 | 91.00 |
| CNN+FastText | 92.64 | 92.12 | 94.60 | 94.01 | 98.63 | 99.00 | 93.78 | 93.00 |
| LSTM+FastText | 90.34 | 90.00 | 91.08 | 90.59 | 96.89 | 97.00 | 92.02 | 92.00 |
| GRU+FastText | 90.10 | 90.00 | 90.49 | 90.80 | 96.98 | 97.00 | 90.79 | 90.62 |
| CNN+Word2Vec | 92.04 | 92.00 | 93.76 | 93.53 | 98.33 | 99.00 | 93.17 | 92.95 |
| LSTM+Word2Vec | 90.19 | 89.86 | 90.00 | 90.00 | 96.15 | 96.00 | 91.50 | 91.00 |
| GRU+Word2Vec | 89.96 | 90.00 | 90.04 | 90.00 | 96.63 | 97.00 | 90.01 | 90.00 |
| **Bangla-BERT** | **96.23** | **96.00** | **92.67** | **93.00** | **99.68** | **99.00** | **96.68** | **97.00** |
| B-ELECTRA | 84.37 | 84.00 | 75.44 | 75.00 | 95.73 | 96.00 | 91.15 | 91.00 |
| M-BERT | 93.38 | 93.00 | 91.06 | 91.00 | 96.93 | 97.00 | 92.13 | 92.00 |
| XML-Roberta | 94.95 | 95.00 | 93.59 | 94.00 | 97.20 | 97.00 | 93.18 | 93.00 |
| Distil-BERT | 93.06 | 93.00 | 89.30 | 89.00 | 94.93 | 95.00 | 92.15 | 92.00 |
| DeBERTa | 94.68 | 95.00 | 78.03 | 78.00 | **99.68** | **99.00** | 95.15 | 95.00 |
| Indic-BERT | 84.30 | 84.00 | 73.30 | 73.00 | 85.86 | 86.00 | 92.33 | 92.00 |
| MuRIL | 95.23 | 95.00 | 78.14 | 78.00 | 99.33 | 99.00 | 96.33 | 96.00 |

i.e., BACC-18, BADD16, BADD6, and LD, discussed in Section 5. The foundation method's performance is discussed, followed by an in-depth analysis of AuthorNet's performance in the subsequent subsection.

### 8.1. Foundation methods performance

Table 4 shows the foundation methods performances, i.e., nine deep learning-based and eight transformer-based models. Three non-contextual embeddings (i.e., GloVe, FastText & Word2Vec) and three deep learning classification methods (i.e., CNN, LSTM & GRU) combinedly produced nine classification models. Among these nine models, CNN+GloVe obtained maximum accuracy of 93.42 for the BACC-18 corpus, 95.03% for the BAAD16 corpus, 98.63% for the LD corpus, and 94.30% for the BAAD6 corpus. In comparison, the minimum accuracies are obtained by the GRU+Word2Vec model for the four corpora. The GRU+Word2Vec model cannot capture the local and global stylometric features due to the lack of syntactic features of the Word2Vec model, and the GRU classifier does not carry the global features.

From Table 4, eight transformer-based models are fine-tuned for four corpora. Among eight models, two are monolingual, i.e., Bengali language specific (Bangla-BERT & B-ELECTRA), and the remaining 6 are multilingual language models. All the models used default hyperparameters, i.e., maximum sequence length 300, batch size 8, and number of epochs 10. Using these default hyperparameters, obtained the maximum accuracy of 96.23% for the BACC-18 corpus, 92.67% for the BAAD16 corpus, 99.68% for the LD corpus, and 96.68% for the BAAD6 corpus. At the same time, the Indic-BERT, B-ELECTRA, and Distil-BERT obtained the minimum performances for four corpora due to a shortage of vocabulary in language models and failed to capture the global stylometric features of author texts (Plaza-del-Arco et al., 2021).

### 8.2. AuthorNet performance

The performance evaluation of AuthorNet is conducted in four different aspects. Firstly, the Hyperparameters Tuning Performance examines the impact of key hyperparameters on the model's performance. Secondly, the Best-performed Model Selection Performance assesses the individual performances of uni-modal models. It selects the best-performing mono-lingual and multi-lingual models for attention-based early fusion. Thirdly, the Attention-based Early Fusion Performances evaluate the performance of the fused models using attention-based mechanisms. Fourthly, Ablation Study, i.e., the impact of different components of AuthorNet. The following paragraph provides a detailed description of each of these evaluation aspects.

#### 8.2.1. Hyperparameters tuning performance

Based on the findings presented in Table 4, the hyperparameter tuning phase of AuthorNet focused on evaluating the performance of 8 different transformer-based language models based on their accuracy (A) and F1-score (F1). Given the specific characteristics and limited vocabulary of Bengali language models, this study placed emphasis on optimizing three crucial hyperparameters: batch size, maximum sequence length, and epoch number. Initial experiments involved varying the batch size among 6, 8, 12, the epoch values among 10, 15, 20, and the maximum sequence lengths among 200, 300, 400. Each transformer-based model produced a total of 27 different models for each corpus, resulting from different combinations of these hyperparameters. Due to time limitations, the hyperparameters were tuned specifically on the BADD16 corpus, and the optimized values were then applied to the remaining corpora (BACC-18, LD, and BADD6). Fig. 4 visually represents the minimum accuracy achieved across the 8 transformer-based models for each of the four corpora. Notably, the minimum accuracies were obtained with 10 epochs, a maximum sequence length of 200, and a batch size of 12.

Based on the observations in Fig. 4, it can be concluded that the performance of the 8 transformer-based models has significantly improved by reducing the batch size from 12 to a smaller value. This adjustment effectively mitigated the issue of gradient overshooting caused by larger batch sizes (i.e., 12) when training on smaller samples. Furthermore, it was observed that the BADD16 corpus exhibited relatively poor performance across the 8 transformer-based models when using a maximum sequence length of 200. This is likely due to the limited availability of stylometric features within the top 200 words of the corpus.

Fig. 5 illustrates the accuracy of the best-performing models among the 8 transformer-based models that were fine-tuned. These models achieved their maximum accuracy when trained for 15 epochs, with a maximum sequence length of 400 and a batch size of 4. In cases where an author's text exceeded 400 words, only the top 400 words were considered, while for shorter texts, padding was applied with zeros.

Fig. 5 depicts the maximum accuracies achieved by the fine-tuned models. The monolingual Bangla-BERT-based models demonstrated the highest accuracies, with 98.45% for BACC-18, 99.42% for BADD16, 99.87% for LD, and 97.67% for BADD6. On the other hand, the multilingual Indic-BERT-based models achieved minimum accuracies of 92.54% for BACC-18, 95.62% for BADD16, 98.3% for LD, and 92.3% for BADD6. It is worth noting that the Bengali-language-based ELECTRA (B-ELECTRA), multilingual Distil-BERT, and Indic-BERT-based models exhibited lower accuracies due to the smaller number of trainable parameters compared to the distributions of the training samples. In conclusion, the monolingual Bangla-BERT-based models outperformed the multilingual models in terms of accuracy.

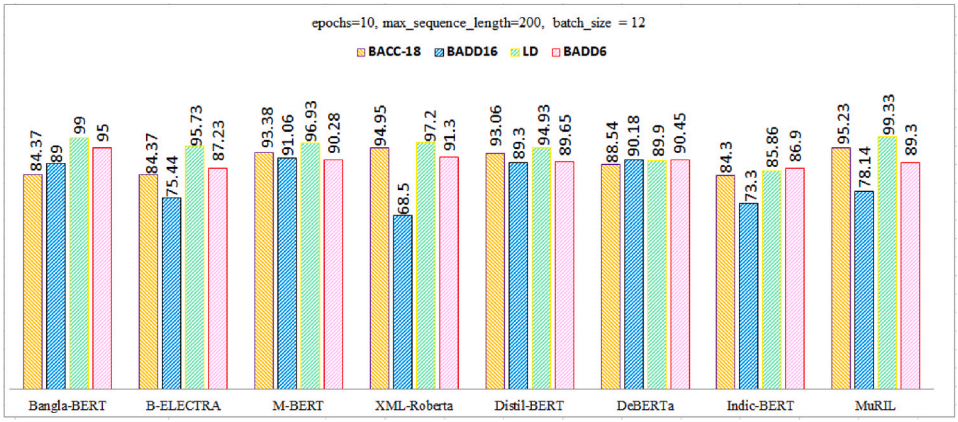



**Fig. 4.** Minimum performed hyperparameters performance of 8 transformer models with 4 corpora.

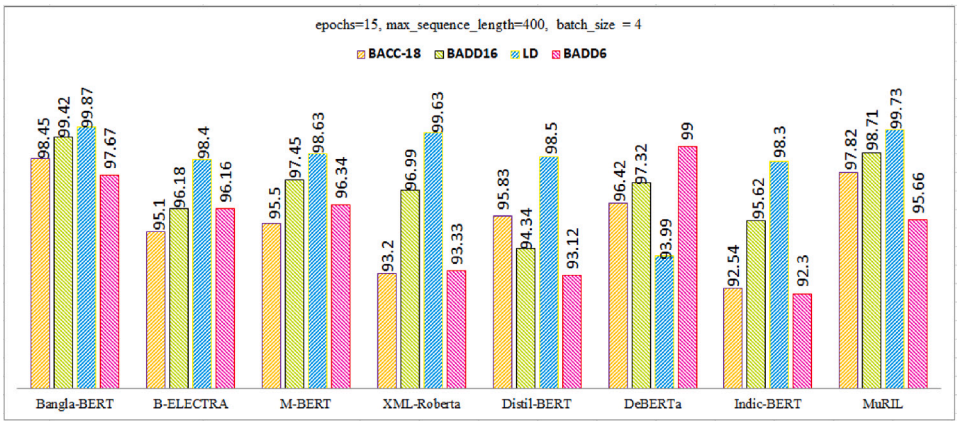



**Fig. 5.** Maximum performed hyperparameters performance of 8 transformer models with 4 corpora.

### *8.2.2. Best-performed model selection performance*

In order to determine the best-performing monolingual and multilingual transformer-based models, this study evaluated the performance of 8 models across four corpora. Table 5 presents the performance metrics, including accuracy (A), F1-score (F1), macro-average recall (mAr), and weighted average recall (wAr), for the four corpora and the eight fine-tuned transformer-based models (Hossain et al., 2024).

In Table 5, the best-performed models are highlighted in bold. The monolingual Bangla-BERT fine-tuned model achieved the highest performance across all four corpora, making it the top-performing model. The second best-performing models were the multilingual fine-tuned MuRIL models, which performed well on three corpora: BACC-18, BADD16, and LD. For the BADD6 corpus, the second-best performance was achieved by the multilingual fine-tuned M-BERT model. The Bengali domain-specific transformers B-ELECTRA did not achieve better accuracy due to limitations in vocabulary and trainable parameters (Hossain et al., 2023). Similarly, DeBERTa, XLM-Roberta, and M-BERT did not achieve better accuracy due to a lack of stylometric features and vocabulary. Overall, the two top-performing models, i.e., monolingual Bangla-BERT and multilingual MuRIL, will be further used in the attention-based early fusion stage.

*Accuracy assessment of input data.* As an illustration, in Table 5, the BAAC-18 corpus test data reveals a maximum accuracy of 98.45% for monolingual Bangla-BERT, with the second-best accuracy at 97.82% for MuRIL. In contrast, other transformer-based models exhibit accuracy ranging from 92.54% to 96.42%. The disparity in accuracy is attributed to the fact that the Bangla-BERT language model is exclusively trained on Bengali, allowing it to extract a multitude of morphological and stylometric features, thereby strengthening its performance. On the other hand, other language models trained on multiple languages include a limited amount of Bengali text, leading to a diminished capability to capture morphological variations, such as (*Sadhu-bhasha and Cholito-bhasa*). Moreover, the accuracy of models is influenced by the distribution of the training and test datasets. For instance, the LD corpus test and training set performances remain closely aligned across all transformer-based language models due to a consistent number of authors and a similar distribution in both datasets.

### *8.2.3. Attention-based early fusion performance*

In this study, we explored two feature-level early fusion techniques: concatenation and averaging. Based on the best-performing models (i.e., Bangla-BERT and MuRIL), we constructed four early fusion-based authorship classification models: concatenation of Bangla-BERT and MuRIL (Cat(.,)), averaging of Bangla-BERT and MuRIL (Avg(.,)), averaging followed by the attention of Bangla-BERT and MuRIL (AAvg(.,)), and concatenation followed by attention-based early fusion of Bangla-BERT and MuRIL (i.e., proposed model). The performance of these attention-based early fusion models can be seen in Table 6.

From Table 6, the proposed AuthorNet, utilizing attention-based early fusion, demonstrated improvements in accuracy compared to the best-performed Bangla-BERT models. Specifically, the proposed model (i.e., AuthorNet) achieved a 0.44% accuracy improvement for the BACC-18 corpus (from 98.45% to 98.86%), 0.07% for the BADD16 corpus (from 99.42% to 99.49%), and 0.24% for the BADD6 corpus (from 97.67% to 97.91%). However, there was no improvement in accuracy for the LD corpus. On the other hand, the average fusion model showed a decrease in accuracy, as it could not effectively capture the complex relationships between the individual models and leverage their respective strengths. In contrast, concatenation-based fusion

**Table 5**
Statistical performance summary of 8 transformer-based fine-tuned models with 4 corpora, Bold values indicate the best and second best-performed models.

| Model | Corpus | A(%) | F1(%) | mAr(%) | wAr(%) |
|---|---|---|---|---|---|
| **Bangla-BERT** | BACC-18 | **98.45 ± 0.01** | **96.00** | **99.00** | **98.00** |
| | BADD16 | **99.42 ± 0.01** | **99.00** | **99.00** | **99.00** |
| | BADD6 | **97.67 ± 0.01** | **98.00** | **98.00** | **98.00** |
| | LD | **99.87 ± 0.01** | **100.00** | **100.00** | **100.00** |
| B-ELECTRA | BACC-18 | 95.10 ± 0.01 | 95.00 | 89.00 | 95.00 |
| | BADD16 | 96.18 ± 0.01 | 96.00 | 95.00 | 96.00 |
| | BADD6 | 96.16 ± 0.01 | 96.00 | 96.00 | 95.00 |
| | LD | 98.40 ± 0.01 | 98.00 | 98.00 | 98.00 |
| M-BERT | BACC-18 | 95.50 ± 0.01 | 96.00 | 95.00 | 96.00 |
| | BADD16 | 97.45 ± 0.01 | 97.00 | 97.00 | 96.00 |
| | BADD6 | **96.34 ± 0.01** | **96.00** | **95.00** | **96.00** |
| | LD | 98.63 ± 0.01 | 99.00 | 99.00 | 98.00 |
| XLM-Roberta | BACC-18 | 93.20 ± 0.01 | 93.00 | 86.00 | 93.00 |
| | BADD16 | 96.99 ± 0.01 | 97.00 | 96.00 | 97.00 |
| | BADD6 | 93.33 ± 0.01 | 98.00 | 98.00 | 98.00 |
| | LD | 99.63 ± 0.01 | 100.00 | 100.00 | 100.00 |
| DistilBERT | BACC-18 | 95.83 ± 0.01 | 96.00 | 96.00 | 96.00 |
| | BADD16 | 94.34 ± 0.01 | 94.00 | 94.00 | 93.00 |
| | BADD6 | 93.12 ± 0.01 | 93.00 | 92.00 | 93.00 |
| | LD | 98.50 ± 0.01 | 99.00 | 99.00 | 100.00 |
| DeBERTa | BACC-18 | 96.42 ± 0.01 | 96.00 | 95.00 | 96.00 |
| | BADD16 | 97.32 ± 0.01 | 97.00 | 96.00 | 96.00 |
| | BADD6 | 93.99 ± 0.01 | 94.00 | 94.00 | 93.00 |
| | LD | 99.00 ± 0.01 | 100.00 | 100.00 | 100.00 |
| Indic-BERT | BACC-18 | 92.54 ± 0.01 | 93.00 | 93.00 | 94.00 |
| | BADD16 | 95.62 ± 0.01 | 96.00 | 96.00 | 97.00 |
| | BADD6 | 92.30 ± 0.01 | 92.00 | 92.00 | 93.00 |
| | LD | 98.3 ± 0.01 | 98.00 | 98.00 | 99.00 |
| **MuRIL** | BACC-18 | **97.82 ± 0.01** | **98.00** | **98.00** | **98.00** |
| | BADD16 | **98.71 ± 0.01** | **99.00** | **99.00** | **99.00** |
| | BADD6 | 95.66 ± 0.01 | 97.00 | 97.00 | 98.00 |
| | LD | **99.73 ± 0.01** | **100.00** | **100.00** | **100.00** |

**Table 6**
Performance summary of attentional and non-attentional early fusion.

| Model | Corpus | A(%) | F1(%) |
|---|---|---|---|
| Cat(Bangla-BERT,MuRIL) | BACC-18 | 98.52 ± 0.01 | 97.00 |
| | BADD16 | 99.47 ± 0.01 | 99.00 |
| | BADD6 | 97.71 ± 0.01 | 98.00 |
| | LD | 99.87 ± 0.01 | 100.00 |
| Avg(Bangla-BERT,MuRIL) | BACC-18 | 98.36 ± 0.01 | 96.00 |
| | BADD16 | 99.21 ± 0.01 | 99.00 |
| | BADD6 | 97.48 ± 0.01 | 97.00 |
| | LD | 99.63 ± 0.01 | 100.00 |
| AAvg(Bangla-BERT,MuRIL) | BACC-18 | 98.43 ± 0.01 | 96.00 |
| | BADD16 | 99.45 ± 0.01 | 99.00 |
| | BADD6 | 97.69 ± 0.01 | 98.00 |
| | LD | 99.87 ± 0.01 | 100.00 |
| **AuthorNet (Proposed)** | **BACC-18** | **98.86 ± 0.01** | **98.00** |
| | **BADD16** | **99.49 ± 0.01** | **99.00** |
| | **BADD6** | **97.91 ± 0.01** | **98.00** |
| | LD | 99.87 ± 0.01 | 100.00 |

successfully incorporated the distinctive features of each model and produced more generalized features. Notably, the attention operation played a crucial role in identifying the important features of each individual model, contributing to the overall performance enhancement of the attention-based early fusion approach (Mehak et al., 2023).

*Value of accuracy for constructed models.* Table 4 illustrates the accuracy values of foundation deep learning and transformer-based models, indicating that the transformer-based models outperform their deep learning models. Consequently, our model construction steps exclusively involved transformer-based language models. Fig. 5 depicts the performances of tuned hyperparameters of transformer-based models, while Table 6 presents the performances of the constructed models across four corpora. For instance, Bangla-BERT achieves accuracies of 98.45% for BACC-18, 96.00% for BADD16, 92.67% for BADD6, and 99.68% for LD, showcasing improvements over the foundation Bangla-BERT model. The maximum accuracy improvement is 5% for the BADD6 corpus, with a minimum improvement of 0.19% for the LD corpus. These enhancements are attributed to the adapted hyperparameters of language models. Having identified the two models with the best accuracy, namely Bangla-BERT, and MuRIL, we subsequently constructed an attention-based early fusion model. The performance of this attention-based early fusion model is summarized in Table 6.

### 8.2.4. Statistical significance test

The statistical hypothesis test known as Welch's t-test was conducted with a significance level of 5% (0.05) to determine whether the performance improvement observed in our proposed model is statistically significant when compared to foundation models (Firdaus et al., 2023; Ramezani, 2021; Welch, 1947). In this study, AuthorNet and eight transformer-based fine-tuned models were executed consecutively six times, and their respective accuracies were recorded. The accuracy values of the foundation models formed one group, while the accuracy of AuthorNet constituted another group. Welch's t-test was then performed to compare these groups, resulting in the calculation of t-values and p-values for four corpora and eight transformer-based models compared to AuthorNet. The summary of the overall t-values and $p$-value can be found in Table 7.

Based on the information presented in Table 7, we can draw the conclusion that the results of the proposed model are statistically significant across four corpora. However, it is worth noting that in the case of the *LD corpus with Bangla-BERT models, the statistical significance compared to the proposed AuthorNet is not observed.*

### 8.2.5. Ablation studies

To address **RQ3**, this study conducted ablation studies in two different scenarios: (i) model-level ablation and (ii) AuthorNet component-level ablation. The model-level ablation analysis aimed to identify the top-performing models among the models evaluated with four corpora. Conversely, the AuthorNet component-level ablation studies investigated the influence of specific components, such as maximum sequence length, dropout values, number of attention layers, and attention layer embedding size, on the accuracy of the AuthorNet model (Mahdaddi et al., 2021).

*Model-level ablation.* Model-level ablation analysis in authorship attribution involves systematically evaluating the performance of different models by selectively disabling or removing specific models and assessing the impact on attribution accuracy. The performance of three deep learning models (CNN, LSTM, and GRU) and eight transformer-based language models (i.e., Bangla-BERT, B-ELECTRA, M-BERT, XML-Roberta, DistilBERT, DeBERTa, Indic-BERT and MuRIL) on four corpora using default hyperparameters is shown in Table 4. The results of the ablation analysis reveal that the transformer-based language models outperformed the deep learning models across all four corpora. In the subsequent phase, ablation analysis was conducted on the eight transformer-based models using tuned hyperparameters for the four corpora, and the findings are summarized in Table 5. Based on the model-level ablation analysis presented in Table Tables 4 and 5, the two best-performing transformer-based models for the four corpora are Bangla-BERT and MuRIL. These models are then used to create attention-based early fusion models, collectively called AuthorNet.

*AuthorNet component-level ablation.* Table 8 provided the results of component-level ablation studies for AuthorNet. These studies analyze the effects of various component configurations (CC), including MSL (Maximum Sequence Length), DV (Dropout Values), NAL (Number of Attention Layers), and ALES (Attention Layer Embedding Size), across four corpora: BACC-18, BADD16, BADD6, and LD.

**Table 7**
Statistical significance test *t*-value and *p*-value summary of transformer-based models with four corpora compared to AuthorNet, Bold values indicate that the difference is statistically significant.

| Model | t-value/p-value | | | |
|---|---|---|---|---|
| | BACC-18 | BADD16 | BADD6 | LD |
| Bangla-BERT | **3.65/0.02** | **11.38/6.3E−06** | **24.00/9.6E−09** | 0.00/1.0 |
| B-ELECTRA | **40.11/1.9E−06** | **523.67/4.7E−18** | **176.00/1.2E−15** | **202.46/3.9E−16** |
| M-BERT | **35.85/3.3E−06** | **254.75/1.7E−13** | **157.00/3.0E−15** | **141.55/8.7E−14** |
| XLM-Roberta | **61.22/3.6E−07** | **473.58/4.4E−19** | **458.00/5.7E−19** | **33.28/7.2E−10** |
| DistilBERT | **32.50/5.0E−06** | **742.76/5.5E−18** | **522.85/7.6E−19** | **156.46/4.2E−14** |
| DeBERTa | **25.88/1.2E−05** | **270.10/1.2E−13** | **23.31/1.9E−05** | **134.71/1.0E−13** |
| Indic-BERT | **68.65/2.4E−07** | **612.53/1.4E−18** | **611.23/2.2E−19** | **179.40/1.5E−14** |
| MuRIL | **10.52/0.00045** | **97.25/6.1E−11** | **225.00/1.7E−16** | **15.37/8.2E−07** |

**Table 8**
AuthorNet component-level ablation analysis summary.

| CC | Values | A(%) | | | |
|---|---|---|---|---|---|
| | | BACC-18 | BADD16 | BADD6 | LD |
| MSL | 100 | 96.65 ± 0.01 | 97.23 ± 0.01 | 95.48 ± 0.01 | 99.23 ± 0.01 |
| | 200 | 97.02 ± 0.01 | 97.83 ± 0.01 | 95.67 ± 0.01 | 99.63 ± 0.01 |
| | 300 | 98.02 ± 0.01 | 98.12 ± 0.01 | 96.07 ± 0.01 | 99.63 ± 0.01 |
| | **400** | **98.86 ± 0.01** | **99.49 ± 0.01** | **97.91 ± 0.01** | **99.87 ± 0.01** |
| DV | 0.50 | 96.65 ± 0.01 | 97.23 ± 0.01 | 95.48 ± 0.01 | 99.23 ± 0.01 |
| | **0.45** | **98.86 ± 0.01** | **99.49 ± 0.01** | **97.91 ± 0.01** | **99.87 ± 0.01** |
| | 0.55 | 97.02 ± 0.01 | 97.83 ± 0.01 | 95.67 ± 0.01 | 99.63 ± 0.01 |
| | 0.60 | 98.02 ± 0.01 | 98.12 ± 0.01 | 96.07 ± 0.01 | 99.63 ± 0.01 |
| NAL | 1 | 96.65 ± 0.01 | 97.23 ± 0.01 | 95.48 ± 0.01 | 99.23 ± 0.01 |
| | **2** | **98.86 ± 0.01** | **99.49 ± 0.01** | **97.91 ± 0.01** | **99.87 ± 0.01** |
| | 3 | 98.02 ± 0.01 | 98.12 ± 0.01 | 96.07 ± 0.01 | 99.63 ± 0.01 |
| | 3 | 97.02 ± 0.01 | 97.83 ± 0.01 | 95.67 ± 0.01 | 99.63 ± 0.01 |
| ALES | 1536 | 96.65 ± 0.01 | 97.23 ± 0.01 | 95.48 ± 0.01 | 99.23 ± 0.01 |
| | **1536/1536** | **98.86 ± 0.01** | **99.49 ± 0.01** | **97.91 ± 0.01** | **99.87 ± 0.01** |
| | 1536/1536/1536 | 98.02 ± 0.01 | 98.12 ± 0.01 | 96.07 ± 0.01 | 99.63 ± 0.01 |
| | 1536/1536/768 | 97.02 ± 0.01 | 97.83 ± 0.01 | 95.67 ± 0.01 | 99.63 ± 0.01 |

From Table 8, the highest accuracies recorded were 98.86% for BACC-18, 99.49% for BADD16, 97.91% for BADD6, and 99.87% for LD. These superior results were achieved using an optimal component configuration consisting of an MSL value of 400, a DV value of 0.45, NAL value of 2, and ALES configuration of 1536/1536. This configuration, featuring a longer maximum sequence length, lower dropout rate, two attention layers, and a specific attention layer embedding size, likely contributed to improved performance by facilitating better generalization, attentiveness, and information representation within the model.

Overall, the model-level ablation analysis provided valuable insights into the effectiveness of different models across the four corpora. Additionally, the component-level analyses of AuthorNet offered comprehensive insights into the performance and configurations of individual components. These findings contribute to a deeper understanding of the underlying factors influencing the accuracy of AuthorNet, ultimately informing the development of more effective and efficient systems for resource-constrained authorship attribution. The results of this study, thus, hold significant implications for advancing the field of low-resource authorship attribution and related research areas.

#### *8.2.6. Impact of out-of-vocabulary and tokenization on AuthorNet accuracy*

Table 9 provides insights into the impact of OOV words and tokenization operations on the accuracy of AuthorNet in authorship attribution. The table includes the number of OOV words (#OOV), the number of tokenized words (#TkW), the number of matched words (#MtW), and the accuracy (A(%)). OOV words represent the words that are not present in the embeddings, while tokenization is applied when an embedding or language model is unable to find a word in the pre-trained vocabulary, breaking the word into sub-words or characters to find a match with embeddings.

The results show that the non-contextual GloVe model has a significant number of OOV words, with 85,342 for BACC-18, 107,804 for BADD16, 47,350 for BADD6, and 13,062 for LD. Similarly, the non-contextual and sub-word-based FastText model has 32,562, 68,310, 16,938, and 5,045 OOV words for the respective corpora. Despite employing tokenization for handling OOV words, these models struggle to achieve higher accuracy compared to AuthorNet, ranging from 92.46% to 98.63%.

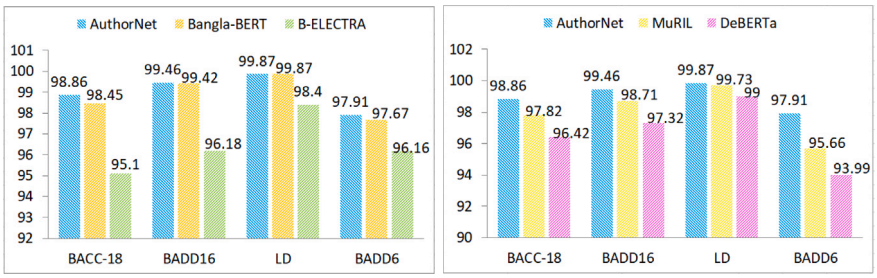
On the other hand, the context-aware models, Bangla-BERT and MuRIL, show no OOV words for any of the corpora, indicating their effectiveness in handling all the words through tokenization. Bangla-BERT achieves accuracies ranging from 97.67% to 99.87%, while MuRIL achieves accuracies ranging from 95.66% to 99.73%. These models highlight the importance of robust language models that capture contextual and semantic features for improved accuracy in authorship attribution tasks.

The proposed AuthorNet, which is an attentional early fusion model, also shows no OOV words across the corpora. This implies that the model can effectively handle the vocabulary present in the datasets. The accuracy of AuthorNet ranges from 97.91% to 99.49%, outperforming the non-contextual embedding and non-fusion models in most cases. This highlights the effectiveness of attentional early fusion and its ability to capture the complex relationships between the models and leverage their strengths for improved accuracy.

Overall, the results suggest that the presence of OOV words can have varying effects on the performance of authorship attribution models. Attentional early fusion models, such as AuthorNet, demonstrate better adaptability to different vocabularies, resulting in higher accuracies. The ability to handle OOV words and perform accurate tokenization is crucial in achieving superior performance in authorship attribution tasks.

#### *8.2.7. Comparison of AuthorNet with monolingual and multilingual approaches*

To assess the effectiveness of the proposed system, this study conducted a comprehensive comparison between AuthorNet and several




(a) Comparison of AuthorNet and Monolingual Best-2 Models Accuracy (b) Comparison of AuthorNet and Multilingual Best-2 Models Accuracy

**Fig. 6.** Monolingual and Multilingual Best-2 Models Accuracy Comparison with the Proposed AuthorNet.

**Table 9**
Impact of OOV words and tokenization operation on AuthorNet accuracy.

| Model | Corpus | #OOV | #TkW | #MtW | A(%) |
|---|---|---|---|---|---|
| GloVe+CNN | BACC-18 | 85,342 | 0 | 294,658 | 93.42 ± 0.01 |
| | BADD16 | 107,804 | 0 | 482,196 | 95.03 ± 0.01 |
| | BADD6 | 47,350 | 0 | 122,650 | 94.30 ± 0.01 |
| | LD | 13,062 | 0 | 66,938 | 98.63 ± 0.01 |
| FastText+CNN | BACC-18 | 32,562 | 52,780 | 294,658 | 92.46 ± 0.01 |
| | BADD16 | 68,310 | 39,494 | 482,196 | 94.60 ± 0.01 |
| | BADD6 | 16,938 | 30,412 | 122,650 | 93.78 ± 0.01 |
| | LD | 5,045 | 8,017 | 66,938 | 98.63 ± 0.01 |
| Bangla-BERT | BACC-18 | 0 | 379,040 | 960 | 98.45 ± 0.01 |
| | BADD16 | 0 | 589,003 | 997 | 99.42 ± 0.01 |
| | BADD6 | 0 | 169,350 | 650 | 97.67 ± 0.01 |
| | LD | 0 | 79,620 | 380 | 99.87 ± 0.01 |
| MuRIL | BACC-18 | 0 | 379,347 | 653 | 97.82 ± 0.01 |
| | BADD16 | 0 | 589,279 | 721 | 98.71 ± 0.01 |
| | BADD6 | 0 | 169,587 | 413 | 95.66 ± 0.01 |
| | LD | 0 | 79,730 | 270 | 99.73 ± 0.01 |
| **AuthorNet** | **BACC-18** | 0 | **758,387** | **1,613** | **98.86 ± 0.01** |
| | **BADD16** | 0 | **1,178,282** | **1,718** | **99.49 ± 0.01** |
| | **BADD6** | 0 | **338,937** | **1,063** | **97.91 ± 0.01** |
| | **LD** | 0 | **159,350** | **650** | **99.87 ± 0.01** |

monolingual and multilingual transformer-based models specifically designed for Bengali authorship attribution. The monolingual models include Bangla-BERT and B-ELECTRA, while the multilingual models comprise M-BERT, XLM-RoBERTa, DistilBERT, DeBERTa, Indic-BERT, and MuRIL. The accuracy results of the two best-performing monolingual models, as well as AuthorNet, are presented in Fig. 6(a), while Fig. 6(b) illustrates the performance of the two selected multilingual models in comparison to AuthorNet. Fig. 6 provides an overall visualization of the accuracy comparisons between AuthorNet and the top-performing monolingual and multilingual models.

From the analysis presented in Fig. 6(a), it is evident that AuthorNet achieves a remarkable accuracy improvement of 3.76% for the BACC-18 corpus compared to the monolingual models. Specifically, AuthorNet achieves an accuracy of 98.86%, while B-ELECTRA attains an accuracy of 95.10%. Similarly, Fig. 6(b) highlights the superior performance of AuthorNet over the multilingual models, with a maximum accuracy improvement of 3.92% for the BADD6 corpus. In this case, AuthorNet achieves an accuracy of 97.91% compared to DeBERTa, which achieves an accuracy of 93.99%. The notable enhancements in accuracy can be attributed to the attention-based early fusion employed by AuthorNet, enabling the capture of fine-grained semantic information at the word level and author-specific patterns at the sentence and paragraph levels. Based on these compelling results, it is evident that the proposed AuthorNet model surpasses the state-of-the-art monolingual and multilingual transformer-based language models, establishing its superiority in Bengali authorship attribution tasks (Kumar et al., 2022).

#### *8.2.8. Comparison of AuthorNet with best-performed foundation and tuned models*

To summarize the performance of the proposed AuthorNet, we conducted a comparative analysis with the best-performing foundation model and the best-performing tuned models. The heatmap in Fig. 7 illustrates the performance of the foundation best-performing model (Basngla-BERT from Table 4), the best-performing tuned model (Bangla-BERT from Table 5), and AuthorNet (from Table 6). Additionally, the heatmap includes the improvement achieved by AuthorNet compared to the best-performing foundation model. The maximum improvement of 7.36% was observed for the BADD16 corpus, while the minimum improvement of 0.19% was noted for the LD corpus.

Adapting the batch size and maximum sequence values played a crucial role in achieving the maximum accuracy for both the tuned model and AuthorNet. The Bangla-BERT language model, trained on an extensive dataset of Bengali language, successfully captured the semantic, syntactic, and context-aware distribution specific to Bengali during the training phase. Consequently, the fine-tuned Bangla-BERT model outperformed other transformer-based models.

#### *8.2.9. Comparison of the proposed AuthorNet with existing approaches*

In this study, we assess the performance of the proposed AuthorNet model in comparison to existing approaches using four benchmark corpora for Bengali authorship attribution: BACC-18, BADD16, BADD6, and LD. Previous studies in this field have employed various traditional feature extraction techniques, including TF-IDF, NB, GloVe, FastText, and Word2Vec. Additionally, both traditional and deep learning-based attribution methods such as SVM, LSTM, ULMFiT, and CNN have been utilized (Khatun et al., 2022, 2019; Lahiri et al., 2017; Shanta et al., 2015; Theophilo et al., 2023). Table 10 presents the accuracy results obtained for the four benchmark corpora using the proposed AuthorNet model and the existing models. The existing models were trained and tested using the default train-test partition and their respective hyperparameters, as mentioned in the Boenninghoff et al. (2019), Khatun et al. (2022, 2019), Lahiri et al. (2017), Shanta et al. (2015), Theophilo et al. (2023). It is noteworthy that the AuthorNet model consistently outperforms the existing approaches across all four corpora.

The proposed AuthorNet model demonstrates significant performance improvements compared to the best-performing existing approach, TL-ULMFiT (Khatun et al., 2022), across four Bengali authorship attribution corpora. Specifically, there is a notable increase of 2.92% for the BACC-18 corpus, 2.58% for the BADD6 corpus, 1.09% for the BADD16 corpus, and 0.24% for the LD corpus. It is worth mentioning that TL-ULMFiT utilized the entire token of the BADD16 and BADD6 corpora, while AuthorNet employed only the first 400 words from each author sample. Despite this difference, the proposed AuthorNet consistently outperforms the previous approaches regarding accuracy for all four corpora. This improvement can be attributed to the early fusion and attentional operations implemented in AuthorNet, which effectively capture author-level stylometric, semantic, and syntactic features necessary for accurate authorship attribution. Based on these results, it can be concluded that AuthorNet surpasses all existing methods in Bengali authorship attribution.

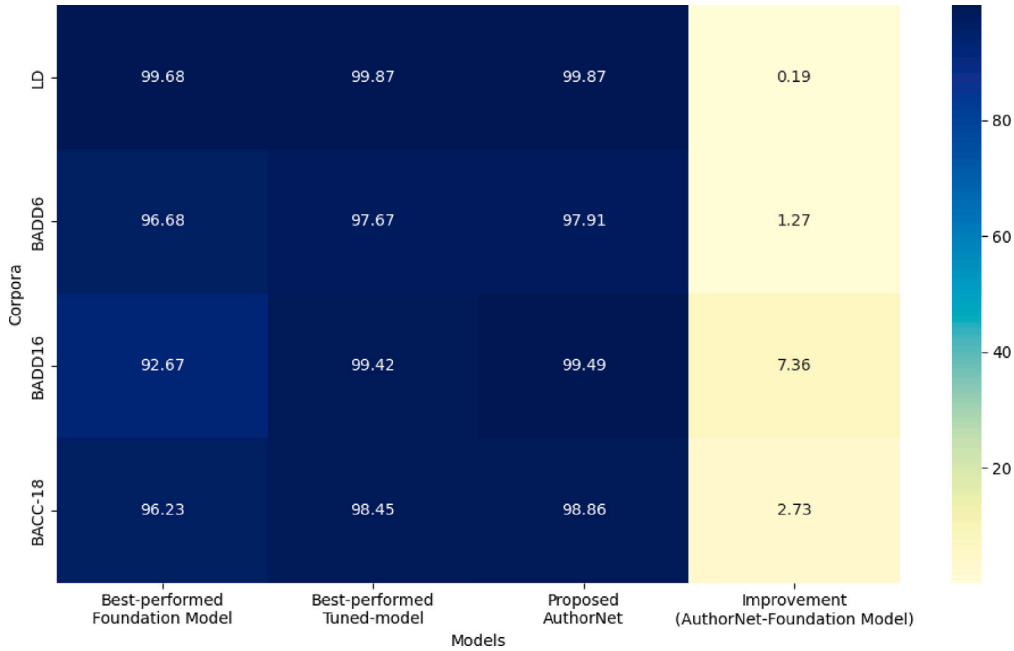



**Fig. 7.** Performance summary of best-performed foundation, tuned, AuthorNet models and improvement of AuthorNet model compared to foundation model.

**Table 10**
Performance compression of proposed AuthorNet and exiting approaches.

| Method | A(%) | | | |
|---|---|---|---|---|
| | BACC-18 | BADD16 | BADD6 | LD |
| **Proposed(AuthorNet)** | **98.86 ± 0.01** | **99.49 ± 0.01** | **97.91 ± 0.01** | **99.87 ± 0.01** |
| GloVe+LSTM (Boenninghoff et al., 2019) | 92.53 ± 0.01 | 93.93 ± 0.01 | 91.45 ± 0.01 | 98.51 ± 0.01 |
| TF+NB (Shanta et al., 2015) | 65.17 ± 0.01 | 62.44 ± 0.01 | 70.45 ± 0.01 | 98.00 ± 0.01 |
| GloVe+SVM (Lahiri et al., 2017) | 68.00 ± 0.01 | 64.21 ± 0.01 | 83.00 ± 0.01 | 94.66 ± 0.01 |
| Fasttext+CNN (Khatun et al., 2019) | 79.44 ± 0.01 | 76.82 ± 0.01 | 75.45 ± 0.01 | 97.60 ± 0.01 |
| TL-ULMFiT (Khatun et al., 2022) | 95.97 ± 0.01 | 98.40 ± 0.01 | 95.33 ± 0.01 | 99.63 ± 0.01 |

### *8.3. Discussion*

This study presents a system for Bengali authorship attribution tasks, utilizing an attention-based early fusion of monolingual and multilingual transformer models. The research highlights the key characteristics and significant findings of this approach as follows:

- In low-resource downstream tasks like Bengali authorship attribution, tuning hyperparameters is a crucial prerequisite. The tuned hyperparameters significantly enhance accuracy, address model overfitting and underfitting issues, and facilitate convergence with a minimum number of epochs. These improvements are demonstrated in Section 8.2.1, along with Figs. 4 and 5.
- In the context of low-resource authorship attribution tasks, the presence of OOV words poses a significant challenge. However, this study tackles this issue by harnessing the power of transformer-based language models. Through this approach, the proposed AuthorNet model outperforms all foundation models by effectively mitigating the impact of OOV words, as demonstrated in Table 9. It is important to note that the tokenization process employed by transformer-based language models may lead to a loss of word-level semantics and syntactic features. However, the overall fine-tuning phase enhances the extraction of author-level stylometric patterns.
- A total of 116 models were constructed using four benchmark corpora in this study. Among these models, 68 were built utilizing three foundation deep-learning classification methods (CNN, LSTM, and GRU) combined with three non-contextual embeddings (GloVe, FastText, and Word2Vec) and eight foundation transformer-based models. Additionally, 32 models were developed using fine-tuned eight transformer-based language models (Bangla-BERT, B-ELECTRA, M-BERT, XLM-RoBERTa, DistilBERT, DeBERTa, Indic-BERT, and MuRIL). In comparison, 16 models were generated through four early fusion methods (concatenation, averaging, concatenation+attention, and averaging+attention). The foundation non-contextual embeddings and deep learning classifiers could not capture context-aware features and adequately handle morphological variations (such as Sadhu-bhasha, Cholito-bhasha, and Sanskrit-bhasha). However, the transformer-based models exhibited improved performance, and the fusion of monolingual and multilingual models yielded even better results.
- **Area of Adequacy for the Constructed Models:** In terms of the area of adequacy, the attention-based early fusion model proves a high applicability to low-resource text classification tasks. This model is particularly useful when default hyperparameters of language models do not perform well for low-resource corpora. It serves as a valuable guide for researchers in the domain of resource-constrained languages, offering insights into constructing well-performing models.

## 9. Error analysis

The error analysis presented in this study provides a comprehensive examination of both the limitations and strengths of the proposed AuthorNet model and the foundation models. Table 11 delineates the error values across four corpora for three distinct phases: foundation models, fine-tuned models, and attention-based early fusion models.

**Table 11**
Error summary of four corpora for Foundation models, Tuned models and Attention-based early fusion models, **bold value indicates the maximum error**.

| Model | Error(%) | | | | |
|---|---|---|---|---|---|
| | BACC-18 | BADD16 | BADD6 | LD | Total |
| Foundation Models | | | | | |
| CNN+GloVe | 6.58 | 4.97 | 1.37 | 5.7 | 18.62 |
| LSTM+GloVe | 8.97 | 7.85 | 2.95 | 7.1 | 26.87 |
| GRU+GloVe | 9.13 | 8.41 | 2.88 | 8.79 | 29.21 |
| CNN+FastText | 7.36 | 5.4 | 1.37 | 6.22 | 20.35 |
| LSTM+FastText | 9.66 | 8.92 | 3.11 | 7.98 | 29.67 |
| GRU+FastText | 9.9 | 9.51 | 3.02 | 9.21 | 31.64 |
| CNN+Word2Vec | 7.96 | 6.24 | 1.67 | 6.83 | 22.7 |
| LSTM+Word2Vec | 9.81 | 10.0 | 3.85 | 8.5 | 32.16 |
| **GRU+Word2Vec** | **10.04** | **9.96** | **3.37** | **9.99** | **33.36** |
| Bangla-BERT | 3.77 | 7.33 | 0.32 | 3.32 | 14.74 |
| B-ELECTRA | 15.63 | 24.56 | 4.27 | 8.85 | 53.31 |
| M-BERT | 6.62 | 8.94 | 3.07 | 7.87 | 26.5 |
| XML-Roberta | 5.05 | 6.41 | 2.8 | 6.82 | 21.08 |
| Distil-BERT | 6.94 | 10.7 | 5.07 | 7.85 | 30.56 |
| DeBERTa | 5.32 | 21.97 | 0.32 | 4.85 | 32.46 |
| **Indic-BERT** | **15.7** | **26.7** | **14.14** | **4.85** | **61.39** |
| MuRIL | 4.77 | 21.86 | 0.67 | 3.67 | 30.97 |
| Tuned Models | | | | | |
| Bangla-BERT | 1.55 | 0.58 | 2.33 | 0.13 | 4.59 |
| B-ELECTRA | 4.90 | 3.82 | 3.84 | 1.60 | 14.16 |
| M-BERT | 4.50 | 2.55 | 3.66 | 1.37 | 12.08 |
| XLM-Roberta | 6.80 | 3.01 | 6.67 | 0.37 | 16.85 |
| DistilBERT | 4.17 | 5.66 | 6.88 | 1.50 | 18.21 |
| DeBERTa | 3.58 | 2.68 | 6.01 | 1.00 | 13.27 |
| **Indic-BERT** | **4.12** | **6.00** | **7.70** | **0.70** | **18.52** |
| MuRIL | 2.18 | 1.29 | 4.34 | 0.27 | 8.08 |
| Attention-based Early Fusion Models | | | | | |
| Cat(Bangla-BERT,MuRIL) | 1.48 | 0.53 | 2.29 | 0.13 | 4.43 |
| **Avg(Bangla-BERT,MuRIL)** | **1.64** | **0.79** | **2.52** | **0.37** | **5.32** |
| AAvg(Bangla-BERT,MuRIL) | 1.57 | 0.55 | 2.31 | 0.13 | 4.56 |
| AuthorNet (Proposed) | 1.14 | 0.51 | 2.09 | 0.13 | 3.87 |

Instances of maximum errors are indicated in bold for each phase. During the initial phase, the foundation models exhibited a maximum total error of 61.39% for the Indic-BERT model and the second-highest error of 33.36% for the GRU+Word2Vec model. The elevated error in Indic-BERT stemmed from challenges in capturing stylometric features within the first 100 words, compounded by inappropriate batch sizes during training, leading to model-level uncertainty. Conversely, the GRU+Word2Vec model encountered data uncertainty and lacked contextual features, particularly when dealing with OOV words. This resulted in the model losing semantic and syntactic meaning at the word level, amplifying attribution errors.

In the subsequent phase, hyperparameter tuning significantly reduced error values across the board, although residual errors persisted. Notably, the maximum error of 18.52% for Indic-BERT was attributed to limitations in stylometric feature extraction and a scarcity of Bengali words in the vocabulary file. The minimal total error of 4.59% for Bangla-BERT was achieved by training the language model exclusively on Bengali, thereby mitigating model-level uncertainties.

In the final phase, the two best-performing models, Bangla-BERT and MuRIL, were selected based on minimal errors, statistical significance tests, and accuracy values. Subsequently, attention-based early fusion models were fine-tuned, reducing errors by integrating features from Bangla-BERT and MuRIL based on attention values. This fusion enhanced stylometric feature representations, culminating in a noteworthy reduction of the overall error. The progression from an initial maximum total error of 61.39% to a second-phase maximum of 18.52% and a final-phase total error of 5.32% underscores the efficacy of tuning and attention-based early fusion in mitigating errors throughout the study.

Fig. 8 offers a visual representation of three author text samples from three different corpora, showcasing the ground truth (i.e., the actual author name), accurately attributed models, and models that failed in attribution (i.e., wrongly attributed models).

From Fig. 8, we can observe three samples: the first sample (S#1) is taken from the BACC-18 corpus, which consists of a poem by "Jibanananda Das". The failure to capture the complete text in the input sample due to sentence segmentation impacts the ability of both the proposed model and the foundation models to capture the actual stylometric patterns.

The second sample (S#2) is extracted from the BADD16 corpus, featuring a story by "Shirshendu Mukhopadhyay". Although the author's text comprises 750 words, the proposed model only considers the first 400 words, resulting in a correct prediction. Conversely, the foundation models fail to provide accurate predictions. This improvement is attributed to the impact of attention-based early fusion operations, which enhance the representation of stylometric features. The third sample (S#3) is derived from the BADD6 corpus, containing an anonymous blog post by "RN". Due to diverse author text feature distribution and morphological variations, such as a mixture of Cholito Bhasha (colloquial language) and transliteration, the proposed AuthorNet model and all of the foundation models struggle to make correct predictions.

*In conclusion, the error analysis reveals that the proposed AuthorNet model has certain limitations.* It faces challenges in providing accurate predictions for sentence-level augmented texts and samples containing morphologically mixed words (including Cholito Bhasha, Sadhu Bhasha, and transliteration). However, when compared to the foundation models, the proposed AuthorNet model demonstrates a lower error rate, indicating its superior performance.

## 10. Conclusion

This study presents a novel authorship attribution system called AuthorNet, which utilizes an attention-based early fusion of monolingual and multilingual transformer-based language models. AuthorNet

| S#. | Input: Transliteration | Ground Truth | Correctly Attributed Models | Wrongly Attributed Models | Corpus:URL |
|---|---|---|---|---|---|
| 1. | শ্রাবণের ... সেই মুখের ভিতর প্রবেশ করলাম: Śrābaṇēra... Sē'i mukhēra bhitara prabēśa karalāma | **Jibanananda Das** | Non of Them | All Baseline Models & AuthorNet | BACC-18: https://tinyurl.com/bddcjw3d |
| 2. | তাহলে ... টেলিগ্রাম করে আমাকে জানাবে তোমার চিঠির: Tāhalē... Ṭēligrāma karē āmākē jānābē tōmāra ciṭhira | **Shirshendu Mukhopadhyay** | AuthorNet | All Baselines Models | BADD16: https://tinyurl.com/3km7xafy |
| 3. | ভালো ... বইটির সাহিত্যমানই আমাকে এই বিশ্বাস দান করেছে: Bhālō ... Ba'iṭira sāhityamāna'i āmākē ē'i biśbāsa dāna karēchē | **RN** | Non of Them | All Baselines & AuthorNet | BADD6: https://tinyurl.com/33brm62a |

**Fig. 8.** Few author text samples with attributed summary.

effectively attributes a given text to a specific author from a predefined list. To enhance the overall authorship attribution accuracy and enable usage in low-resource downstream tasks, we have also developed an automatic hyperparameter tuning algorithm specifically designed for fine-tuning transformer-based language models (i.e., **ARQ1**). This algorithm not only improves the performance of AuthorNet but also proves useful for various other tasks with limited resources. One of the key strengths of AuthorNet is its ability to address OOV issues by leveraging both monolingual and multilingual transformer-based language models (i.e., **ARQ1**).

Additionally, this study incorporated an attention-based early fusion of monolingual and multilingual transformer-based language models, effectively integrating their distinct features to enhance overall performance (i.e., **ARQ2**). We examined the impact of model-level and AuthorNet component-level factors on authorship attribution tasks (**ARQ3**). Notably, this study stands as the first to develop an attention-based early fusion system for authorship attribution, which underwent comprehensive evaluation across four benchmark corpora and was compared against three deep learning and eight transformer-based fine-tuned classification methods. The evaluation results unequivocally demonstrate that the proposed AuthorNet surpasses all foundation models in terms of performance.

*Future works.* For future research, there are several promising directions to explore based on the findings of this study. Considering the success of AuthorNet, further improvements can be achieved by incorporating large language model (LLM), pre-trained transformer-based language models specifically tailored for Bengali, such as DeBERTa, ELECTRA, and BERT. This would enhance the quality of sub-word or character-level features in the language models and potentially lead to even higher accuracy in authorship attribution tasks. Investigate the adaptability of AuthorNet to multilingual and Code-mixed data authorship attribution tasks. Assess the model's performance when applied to texts in multiple languages, particularly those with linguistic similarities or influences on Bengali. Enhance the interpretability and explainability of AuthorNet's decisions leading to explainable LLMs. Conduct a comprehensive analysis of the model's ethical considerations, addressing potential biases in the training data and model predictions. Implement measures to mitigate biases and ensure fair treatment across diverse demographic groups.

## CRediT authorship contribution statement

**Md. Rajib Hossain:** Conceptualization, Data curation, Methodology, Writing – original draft, Software. **Mohammed Moshiul Hoque:** Supervision, Conceptualization, Writing – review & editing. **M. Ali Akber Dewan:** Software, Validation, Editing. **Enamul Hoque:** Writing – review & editing. **Nazmul Siddique:** Writing – review & editing, Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

## References

Al-Sarem, M., Saeed, F., Alsaeedi, A., Boulila, W., & Al-Hadhrami, T. (2020). Ensemble methods for instance-based arabic language authorship attribution. *IEEE Access*, *8*, 17331–17345. http://dx.doi.org/10.1109/ACCESS.2020.2964952.

Alam, T., Khan, A., & Alam, F. (2020). Bangla text classification using transformers. CoRR, arXiv:2011.04446.

Anisuzzaman, D. M., & Salam, A. (2018). Authorship attribution for Bengali language using the fusion of N-Gram and Naive Bayes algorithms. *International Journal of Information Technology and Computer Science*, *10*, 11–21. http://dx.doi.org/10.5815/ijitcs.2018.10.02.

Anwar, W., Bajwa, I. S., Choudhary, M. A., & Ramzan, S. (2018). An empirical study on forensic analysis of urdu text using LDA-based authorship attribution. *IEEE Access*, *7*, 3224–3234. http://dx.doi.org/10.1109/ACCESS.2018.2885011.

Bander, A., Edwin, D., Richard, H., Spiros, M., & Rachel, G. (2017). Source code authorship attribution using long short-term memory based networks. In *Proc. ESORICS 2017* (pp. 65–82). Oslo, Norway.

Bhattacharjee, A., Hasan, T., Samin, K., Rahman, M. S., Iqbal, A., & Shahriyar, R. (2021). BanglaBERT: Combating embedding barrier for low-resource language understanding. CoRR, arXiv:2101.00204.

Boenninghoff, B. T., Nickel, R. M., Zeiler, S., & Kolossa, D. (2019). Similarity learning for authorship verification in social media. CoRR, arXiv:1908.07844.

Borroto, M. A., & Ricca, F. (2023). SPARQL-QA-v2 system for knowledge base question answering. *Expert Systems with Applications*, *229*, Article 120383. http://dx.doi.org/10.1016/j.eswa.2023.120383.

Chakraborty, T. (2012). Authorship identification using stylometry analysis in Bengali literature. CoRR, arXiv:1208.6268.

Chen, X., Wang, H., Ke, L., Lu, Z., Su, H., & Chen, X. (2023). Identifying Cantonese rumors with discriminative feature integration in online social networks. *Expert Systems with Applications*, *215*, Article 119347. http://dx.doi.org/10.1016/j.eswa.2022.119347.

Chowdhury, H. A., Imon, M. A. H., & Islam, M. S. (2018). A comparative analysis of word embedding representations in authorship attribution of bengali literature. In *2018 21st international conference of computer and information technology* (pp. 1–6). http://dx.doi.org/10.1109/ICCITECHN.2018.8631977.

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th annual meeting of the association for computational linguistics* (pp. 8440–8451). Online: Association for Computational Linguistics, http://dx.doi.org/10.18653/v1/2020.acl-main.747.

Das, P., Tasmim, R., & Ismail, S. (2015). An experimental study of stylometry in Bangla literature. In *Proc. EICT* (pp. 575–580). Cox's Bazar, Bangladesh.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)* (pp. 4171–4186). Minneapolis, Minnesota: Association for Computational Linguistics, http://dx.doi.org/10.18653/v1/N19-1423.

Dipongkor, A. K., Islam, M. S., Kayesh, H., Hossain, M. S., Anwar, A., Rahman, K. A., & Razzak, I. (2021). DAAB: Deep authorship attribution in Bengali. In *2021 international joint conference on neural networks* (pp. 1–9). http://dx.doi.org/10.1109/IJCNN52387.2021.9533619.

Fabien, M., Villatoro-Tello, E., Motlicek, P., & Parida, S. (2020). BertAA : BERT fine-tuning for authorship attribution. In *Proceedings of the 17th international conference on natural language processing* (pp. 127–137). Indian Institute of Technology Patna, Patna, India: NLP Association of India (NLPAI).

Firdaus, M., Ekbal, A., & Cambria, E. (2023). Multitask learning for multilingual intent detection and slot filling in dialogue systems. *Information Fusion*, *91*, 299–315. http://dx.doi.org/10.1016/j.inffus.2022.09.029.

Gheewala, S., Xu, S., Yeom, S., & Maqsood, S. (2024). Exploiting deep transformer models in textual review based recommender systems. *Expert Systems with Applications*, *235*, Article 121120. http://dx.doi.org/10.1016/j.eswa.2023.121120.

He, P., Liu, X., Gao, J., & Chen, W. (2021). Deberta: decoding-enhanced bert with disentangled attention. In *International conference on learning representations*. URL https://openreview.net/forum?id=XPZIaotutsD.

Hossain, M. R., & Hoque, M. M. (2018). Automatic Bengali document categorization based on word embedding and statistical learning approaches. In *Proc. IC4ME2* (pp. 1–6). Rajshahi, Bangladesh: http://dx.doi.org/10.1109/IC4ME2.2018.8465632.

Hossain, M. R., & Hoque, M. M. (2023). CoBertTC: Covid-19 text classification using transformer-based language models. In *Intelligent computing and optimization* (pp. 179–186). Cham: Springer Nature Switzerland.

Hossain, M. R., Hoque, M. M., Dewan, M. A. A., Siddique, N., Islam, N., & Sarker, I. H. (2021). Authorship classification in a resource constraint language using convolutional neural networks. *IEEE Access*, *9*, 100319–100338. http://dx.doi.org/10.1109/ACCESS.2021.3095967.

Hossain, M. R., Hoque, M. M., & Siddique, N. (2023). Leveraging the meta-embedding for text classification in a resource-constrained language. *Engineering Applications of Artificial Intelligence*, *124*, Article 106586. http://dx.doi.org/10.1016/j.engappai.2023.106586.

Hossain, M. R., Hoque, M. M., Siddique, N., & Dewan, M. A. A. (2024). AraCovTexFinder: Leveraging the transformer-based language model for Arabic COVID-19 text identification. *Engineering Applications of Artificial Intelligence*, *133*, Article 107987. http://dx.doi.org/10.1016/j.engappai.2024.107987.

Hossain, M. R., Hoque, M. M., Siddique, N., & Sarker, I. H. (2021). Bengali text document categorization based on very deep convolution neural network. *Expert Systems with Applications*, *184*, Article 115394. http://dx.doi.org/10.1016/j.eswa.2021.115394.

Hossain, M. T., Rahman, M. M., Ismail, S., & Islam, M. S. (2017). A stylometric analysis on Bengali literature for authorship attribution. In *Porc. ICCIT* (pp. 1–5). Dhaka, Bangladesh.

Islam, N., Hoque, M. M., & Hossain, M. R. (2017). Automatic authorship detection from Bengali text using stylometric approach. In *Proc. ICCIT* (pp. 1–6). Dhaka, Bangladesh.

Jambi, K. M., Khan, I. H., Siddiqui, M. A., & Alhaj, S. O. (2021). Towards authorship attribution in arabic short-microblog text. *IEEE Access*, *9*, 128506–128520. http://dx.doi.org/10.1109/ACCESS.2021.3112624.

Kabala, J. (2018). Computational authorship attribution in medieval Latin corpora: the case of the Monk of Lido (ca. 1101–08) and Gallus Anonymous (ca. 1113–17). *Language Resources and Evaluation*, *54*, 25–56.

Kakwani, D., Kunchukuttan, A., Golla, S., N.C., G., Bhattacharyya, A., Khapra, M. M., & Kumar, P. (2020). IndicNLPSuite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for Indian languages. In *Findings of the association for computational linguistics: EMNLP 2020* (pp. 4948–4961). Online: Association for Computational Linguistics, http://dx.doi.org/10.18653/v1/2020.findings-emnlp.445.

Khanuja, S., Bansal, D., Mehtani, S., Khosla, S., Dey, A., Gopalan, B., Margam, D. K., Aggarwal, P., Nagipogu, R. T., Dave, S., Gupta, S., Gali, S. C. B., Subramanian, V., & Talukdar, P. P. (2021). MuRIL: Multilingual representations for Indian languages. CoRR, arXiv:2103.10730.

Khatun, A., Rahman, A., Islam, M. S., Chowdhury, H. A., & Tasnim, A. (2022). Authorship Attribution in Bangla Literature (AABL) via transfer learning using ULMFiT. *ACM Transactions on Asian Low-Resource Language Information Processing*, http://dx.doi.org/10.1145/3530691.

Khatun, A., Rahman, A., Islam, M. S., & Marium-E-Jannat (2019). Authorship attribution in Bangla literature using character-level CNN. In *Proc. ICCIT* (pp. 1–5). Dhaka, Bangladesh.

Kumar, A., Pratap, A., Singh, A. K., & Saha, S. (2022). Addressing domain shift in neural machine translation via reinforcement learning. *Expert Systems with Applications*, *201*, Article 117039. http://dx.doi.org/10.1016/j.eswa.2022.117039.

Lahiri, S., Phani, S., & Biswas, A. (2016). A supervised authorship attribution framework for Bengali language. CoRR, arXiv:1607.05650.

Lahiri, S., Phani, S., & Biswas, A. (2017). A supervised learning approach for authorship attribution of bengali literary texts. *ACM Transactions on Asian Low-Resource Language Information Processing*, *16*(4).

Luyckx, K., Daelemans, W., & Vanhoutte, E. (2006). Stylogenetics: Clustering-based stylistic analysis of literary corpora. In *Proc. lREC'06* (pp. 30–35). Genoa, Italy.

Mahdaddi, A., Meshoul, S., & Belguidoum, M. (2021). EA-based hyperparameter optimization of hybrid deep learning models for effective drug-target interactions prediction. *Expert Systems with Applications*, *185*, Article 115525. http://dx.doi.org/10.1016/j.eswa.2021.115525.

Mehak, G., Muneer, I., & Nawab, R. M. A. (2023). Urdu text reuse detection at phrasal level using sentence transformer-based approach. *Expert Systems with Applications*, *234*, Article 121063. http://dx.doi.org/10.1016/j.eswa.2023.121063.

Mutlu, B., & Sezer, E. A. (2023). Enhanced sentence representation for extractive text summarization: Investigating the syntactic and semantic features and their contribution to sentence scoring. *Expert Systems with Applications*, *227*, Article 120302. http://dx.doi.org/10.1016/j.eswa.2023.120302.

Pal, U., Nipu, A. S., & Ismail, S. (2017). A machine learning approach for stylometric analysis of bangla literature. In *Proc. ICCIT* (pp. 1–5). Dhaka, Bangladesh.

Plaza-del-Arco, F. M., Molina-González, M. D., Ureña-López, L. A., & Martín-Valdivia, M. T. (2021). Comparing pre-trained language models for Spanish hate speech detection. *Expert Systems with Applications*, *166*, Article 114120. http://dx.doi.org/10.1016/j.eswa.2020.114120.

Prabhakar, S. K., & Lee, S.-W. (2023). Holistic approaches to music genre classification using efficient transfer and deep learning techniques. *Expert Systems with Applications*, *211*, Article 118636. http://dx.doi.org/10.1016/j.eswa.2022.118636.

Rakshit, G., Ghosh, A., Bhattacharyya, P., & Haffari, G. (2015). Automated analysis of bangla poetry for classification and poet identification. In *Proc. ICON* (pp. 247–253). Trivandrum, India.

Ramezani, R. (2021). A language-independent authorship attribution approach for author identification of text documents. *Expert Systems with Applications*, *180*, Article 115139. http://dx.doi.org/10.1016/j.eswa.2021.115139.

Rocha, A., Scheirer, W. J., Forstall, C. W., Cavalcante, T., Theophilo, A., Shen, B., Carvalho, A. R. B., & Stamatatos, E. (2017). Authorship attribution for social media forensics. *IEEE Transactions on Information Forensics and Security*, *12*(1), 5–33. http://dx.doi.org/10.1109/TIFS.2016.2603960.

Rocha, A., Scheirer, W. J., Forstall, C. W., Cavalcante, T., Theophilo, A., Shen, B., Carvalho, A. R. B., & Stamatatos, E. (2017). Authorship attribution for social media forensics. *IEEE Transactions on Information Forensics and Security*, *12*(1), 5–33. http://dx.doi.org/10.1109/TIFS.2016.2603960.

Ruder, S., Ghaffari, P., & Breslin, J. G. (2016). Character-level and multi-channel convolutional neural networks for large-scale authorship attribution. CoRR, arXiv:1609.06686.

Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. CoRR, arXiv:1910.01108.

Shanta, P., Shibamouli, L., & Arindam, B. (2015). Authorship attribution in Bengali language. In *Proc. ICON* (pp. 100–105). Trivandrum, India.

Suman, C., Raj, A., Saha, S., & Bhattacharyya, P. (2021). Authorship attribution of microtext using capsule networks. *IEEE Transactions on Computational Social Systems*, 1–10. http://dx.doi.org/10.1109/TCSS.2021.3067736.

Taha, A., Khalil, H. M., & El-shishtawy, T. (2021). A two level learning model for authorship authentication. *PLOS ONE*, *16*(8), 1–16. http://dx.doi.org/10.1371/journal.pone.0255661.

Tamboli, M. S., & Prasad, R. S. (2013). Authorship analysis and identification techniques: A review. *International Journal of Computer Applications*, *77*, 11–15. http://dx.doi.org/10.5120/13566-1375.

Theophilo, A., Giot, R., & Rocha, A. (2023). Authorship attribution of social media messages. *IEEE Transactions on Computational Social Systems*, *10*(1), 10–23. http://dx.doi.org/10.1109/TCSS.2021.3123895.

Theophilo, A., & Rocha, A. (2022). Authorship attribution of small messages through language models. In *2022 IEEE international workshop on information forensics and security* (pp. 1–6). http://dx.doi.org/10.1109/WIFS55849.2022.9975413.

Tweedie, F. J., Singh, S., & Holmes, D. I. (1996). Neural network applications in stylometry: The federalist papers. *Computers & Humanities*, *30*(1), 1–10. http://dx.doi.org/10.1007/BF00054024.

Welch, B. L. (1947). The generalization of 'student's' problem when several different population varlances are involved. *Biometrika*, *34*(1–2), 28–35. http://dx.doi.org/10.1093/biomet/34.1-2.28.

Wu, S., & Dredze, M. (2020). Are all languages created equal in multilingual BERT? In *Proceedings of the 5th workshop on representation learning for NLP* (pp. 120–130). Online: Association for Computational Linguistics, http://dx.doi.org/10.18653/v1/2020.repl4nlp-1.16.

Yang, X., Xu, G., Li, Q., Guo, Y., & Zhang, M. (2017). Authorship attribution of source code by using back propagation neural network based on particle swarm optimization. *PLoS One*, *12*, 1–18. http://dx.doi.org/10.1371/journal.pone.0187204.